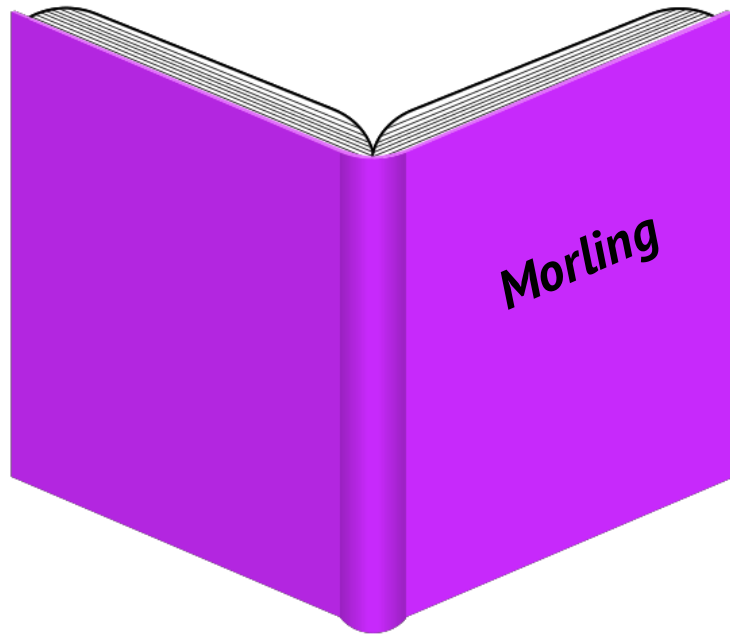




Research Methods and Statistics

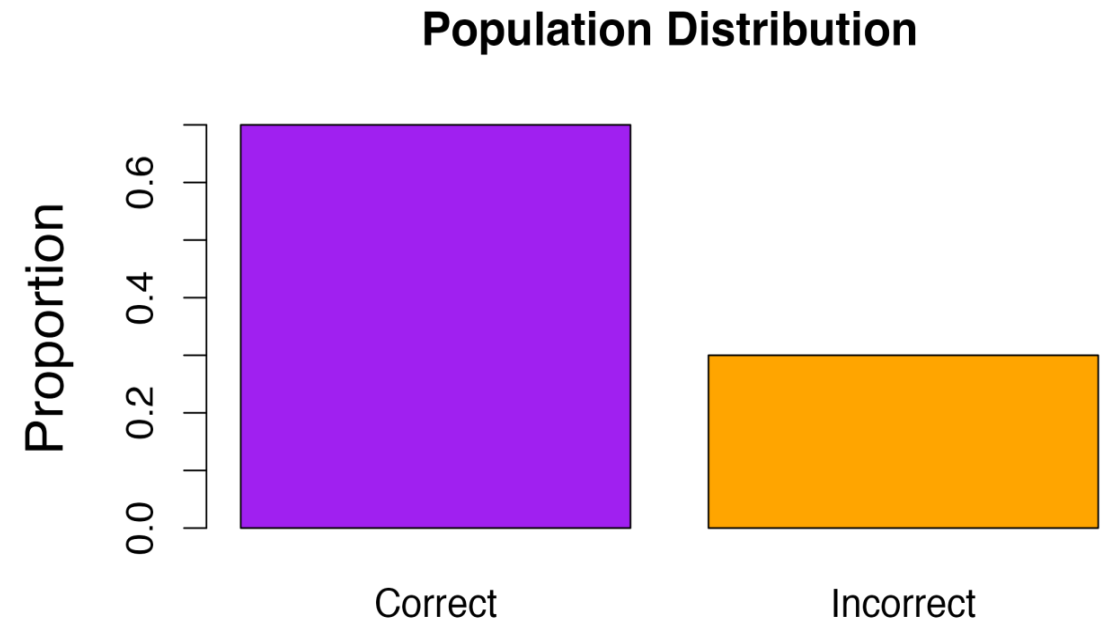
Lecture 10: Sampling Distributions

Johnny van Doorn

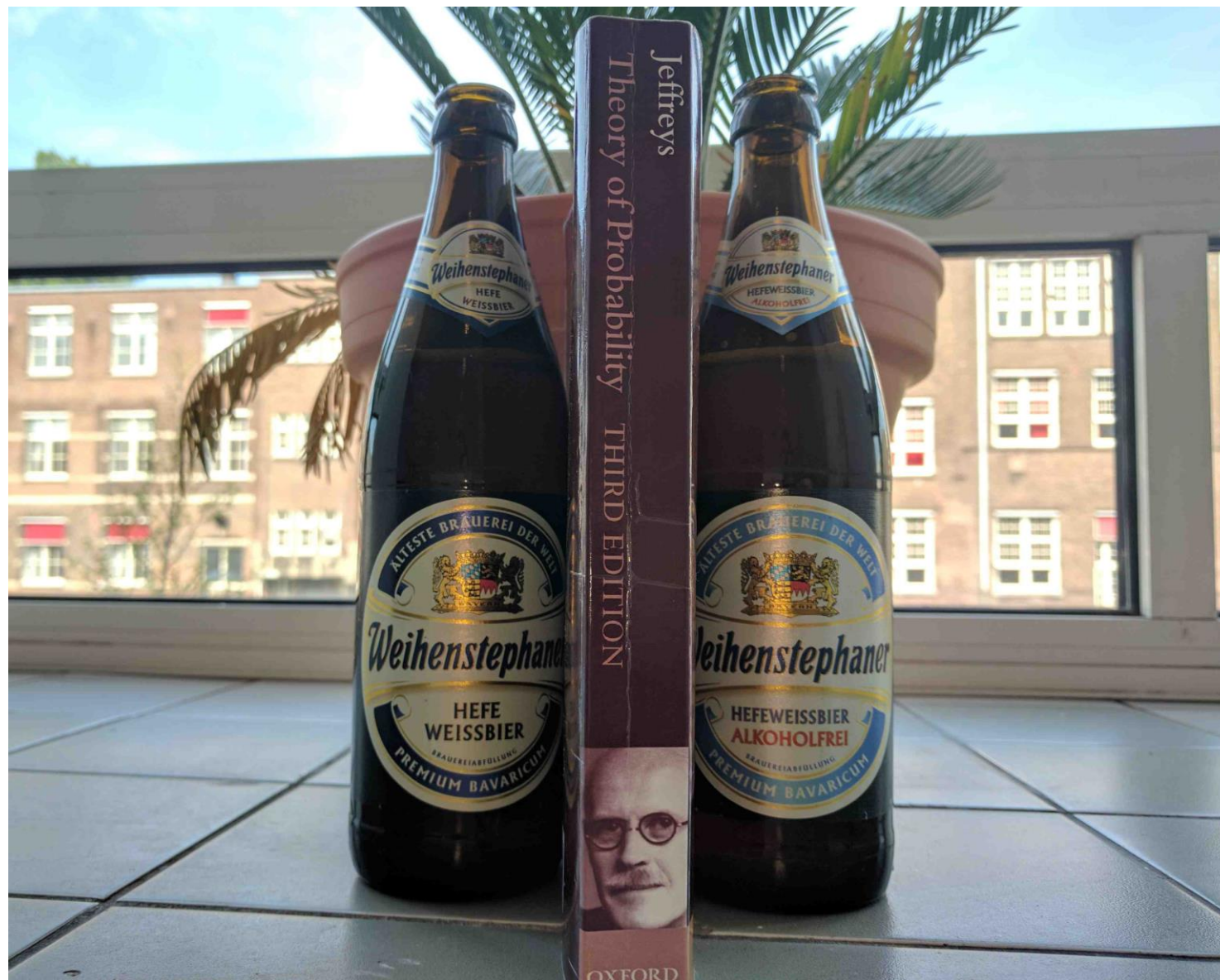


Pictures source: pixabay

Case 1: Beer tasting



- Can people taste the difference between alcoholic and non-alcoholic beer?

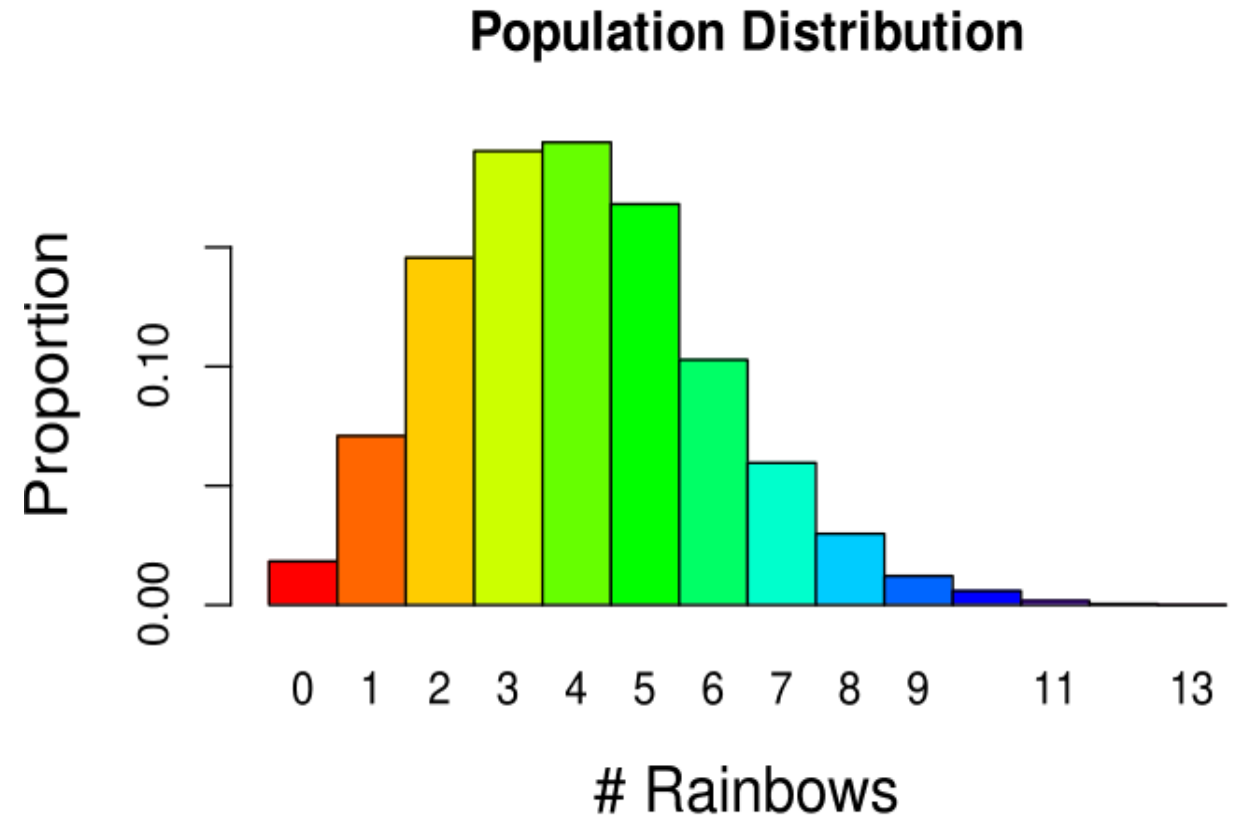






Case 2: Rainbows

- How many rainbows have you seen this year?



Overview of Today

1. **Population distribution vs data distribution**
2. Sampling distribution
 1. Distribution of a quantitative variable
 2. Distribution of a binary categorical variable
3. Central Limit Theorem
4. Excel & Demo
5. Recap
 - Next time
 - Example exam questions

What is a sampling distribution?!

How can we use it for statistical inference?

Why are statistics needed again (Lecture 3)?

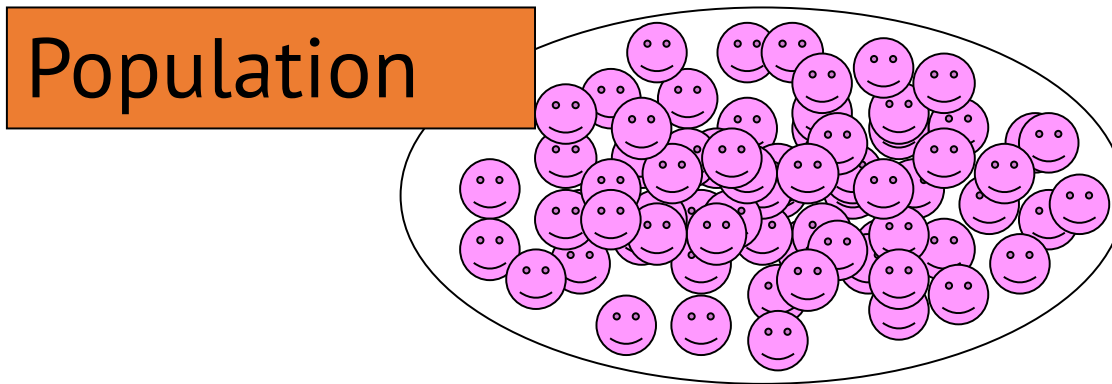
- Inferential statistics (vs. descriptive statistics)
 - Making predictions and decisions about the population

Population: **All** scores/data we are interested in

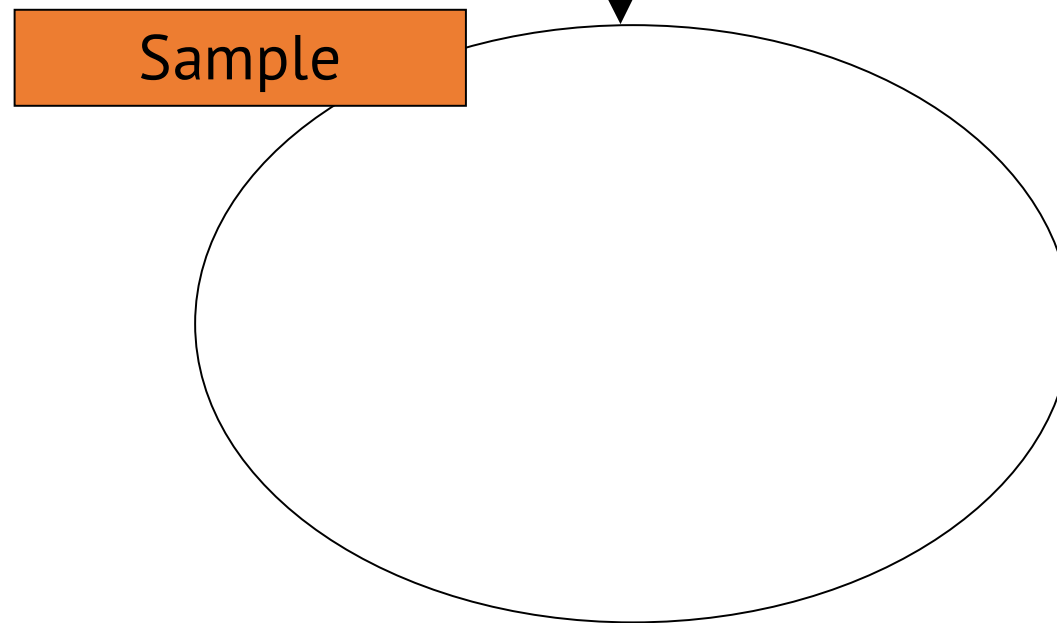
Sample: The part of the population that we have actually observed

Inference: Drawing a conclusion about the population, based on the sample

- Many things can go wrong when drawing inferences



Parameter



Statistic

Statistic vs parameter

A statistic: A numerical summary of the **data** is called a statistic

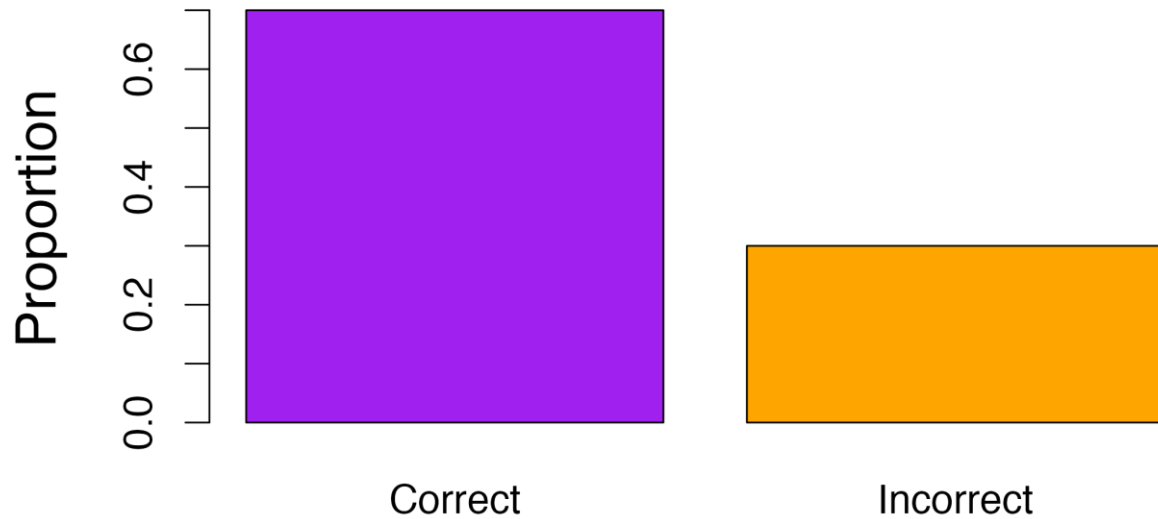
- You can compute this! (mean, median, standard deviation, etc)

A parameter: A numerical summary of the **population** is called a parameter

- This cannot be computed (usually)
- ⑦ **We use statistics to estimate parameters**
- *Sampling distribution* tells us something about the uncertainty of that estimate

Population distribution

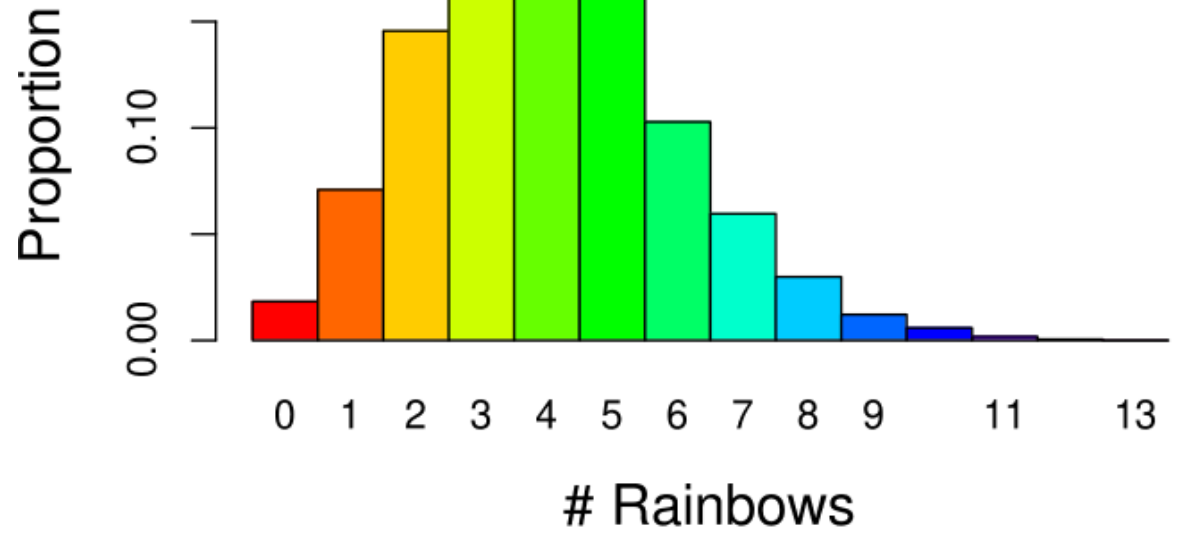
Population Distribution



$$p = 0.7$$

Parameter

Population Distribution

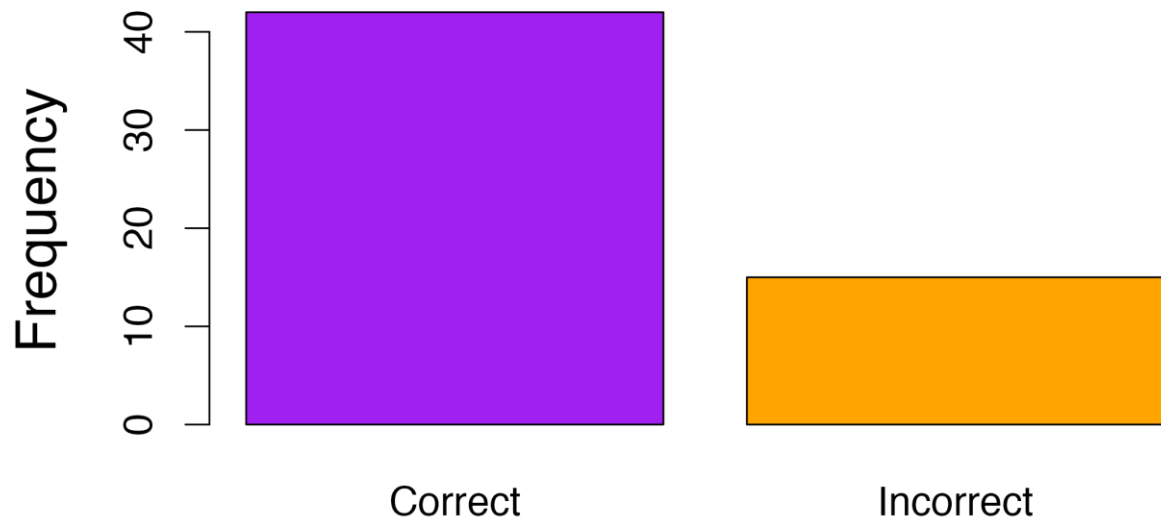


$$\mu = 4$$

Data distribution

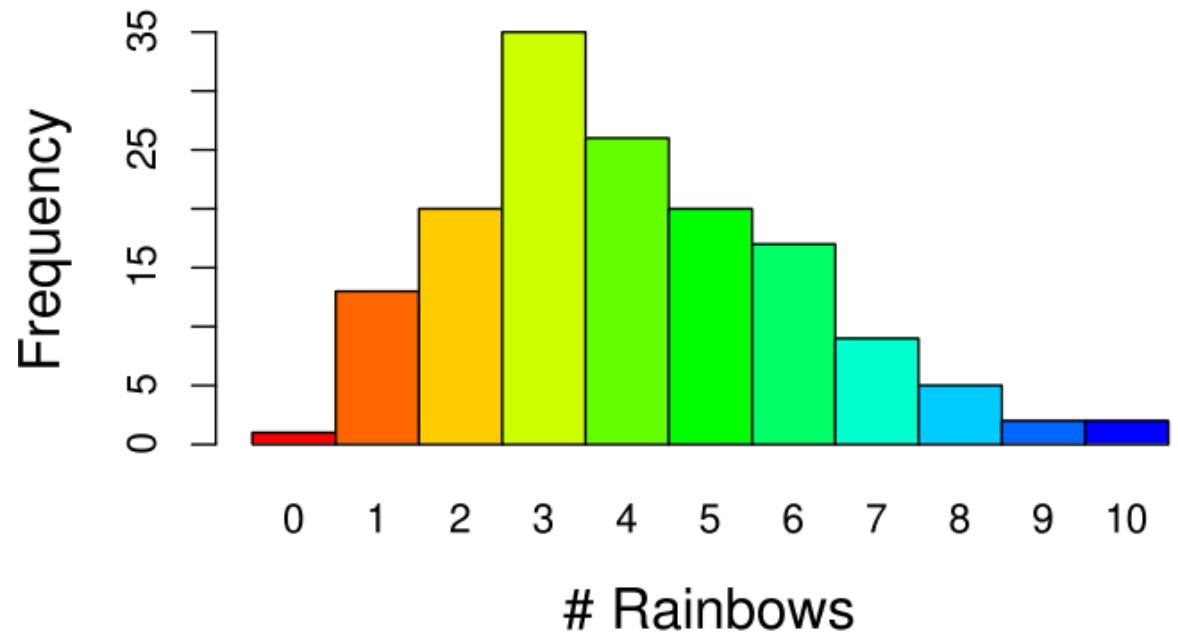
$n = 57$

Data Distribution



$n = 150$

Data Distribution



Note the symbol ("p-hat"). The book does not introduce this until Ch 8.

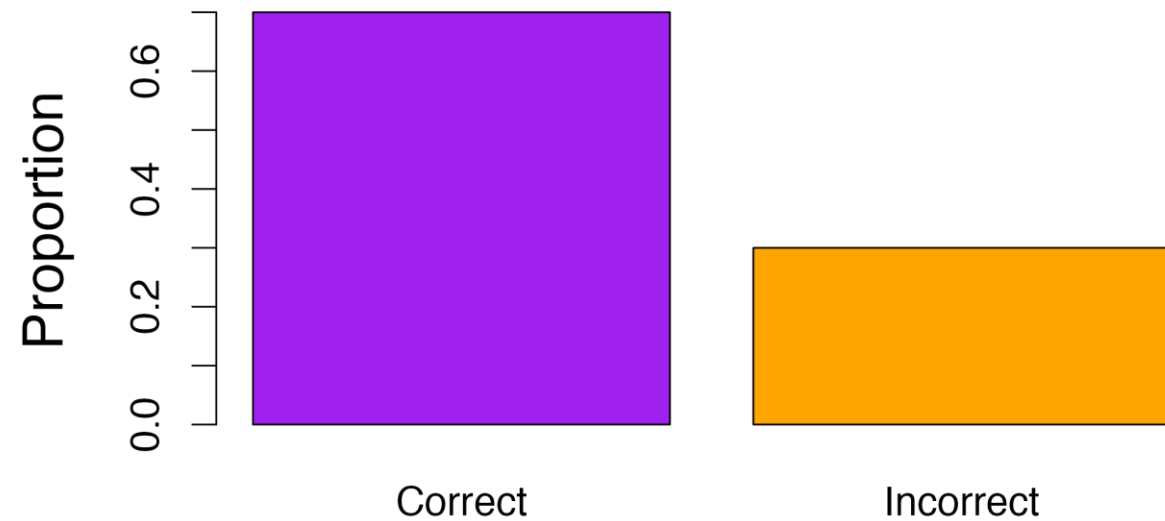
$\hat{p} = 0.74$

Statistic

$\bar{x} = 3.89$

Population and data distribution

Population Distribution

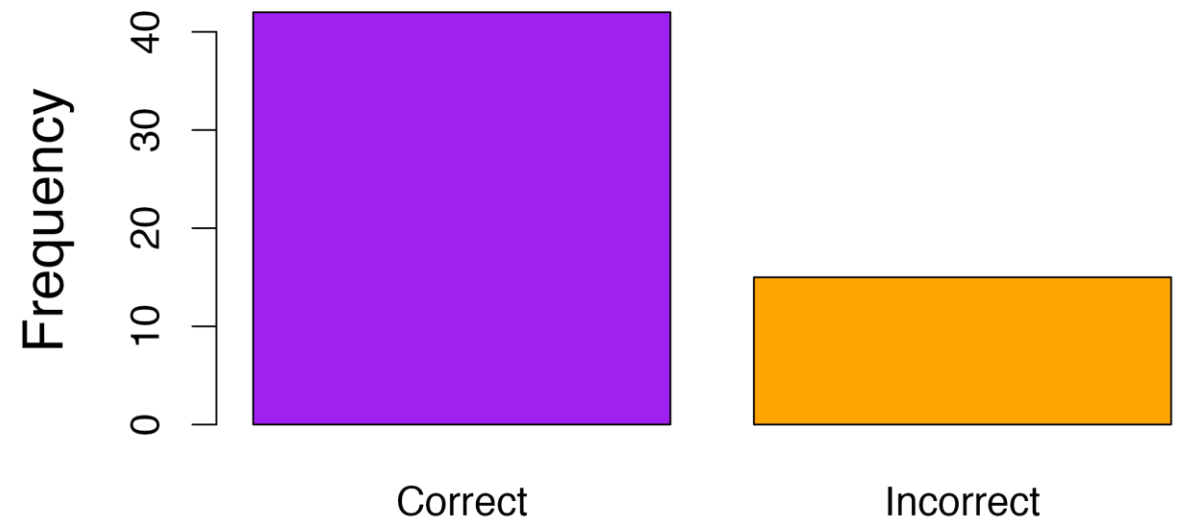


Parameter

$$p = 0.7$$

$n = 57$

Data Distribution

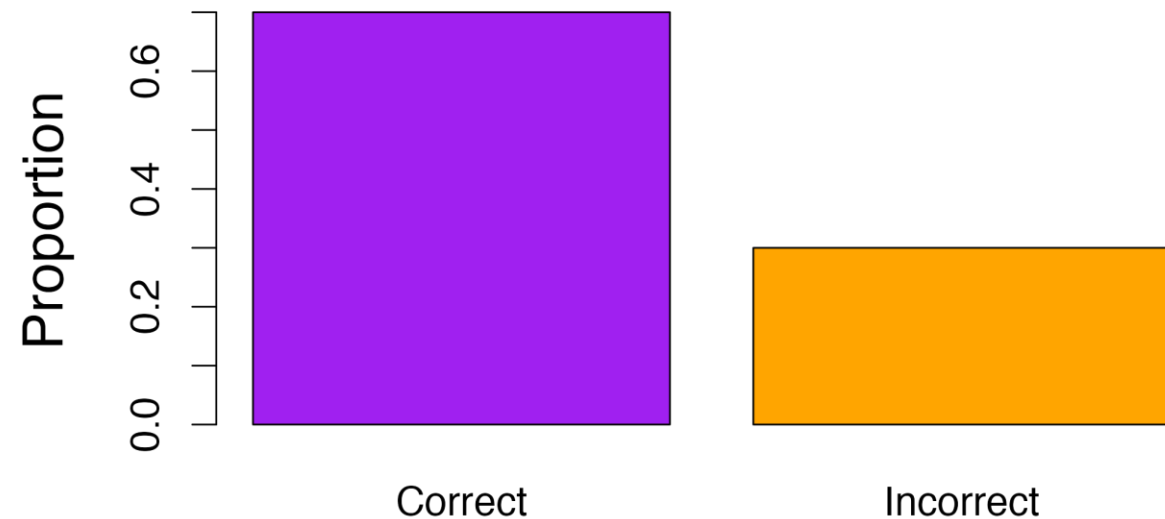


Statistic

$$\hat{p} = 0.74$$

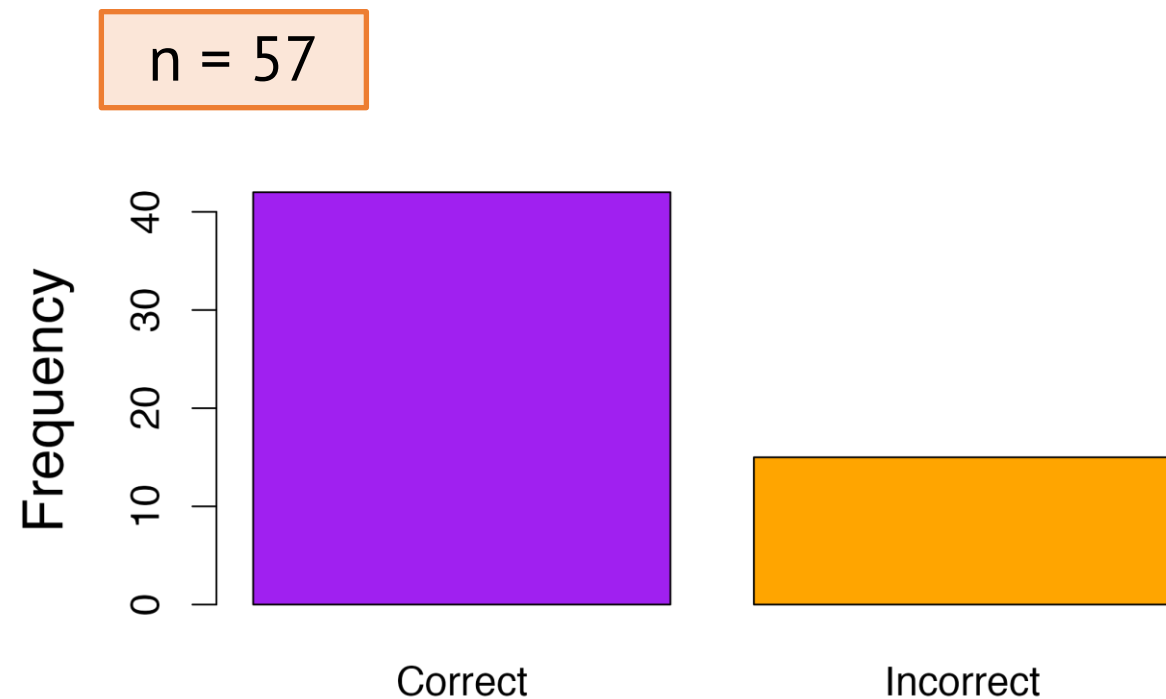
Population and data distribution

Spot the 3 differences!



Parameter

$$p = 0.7$$



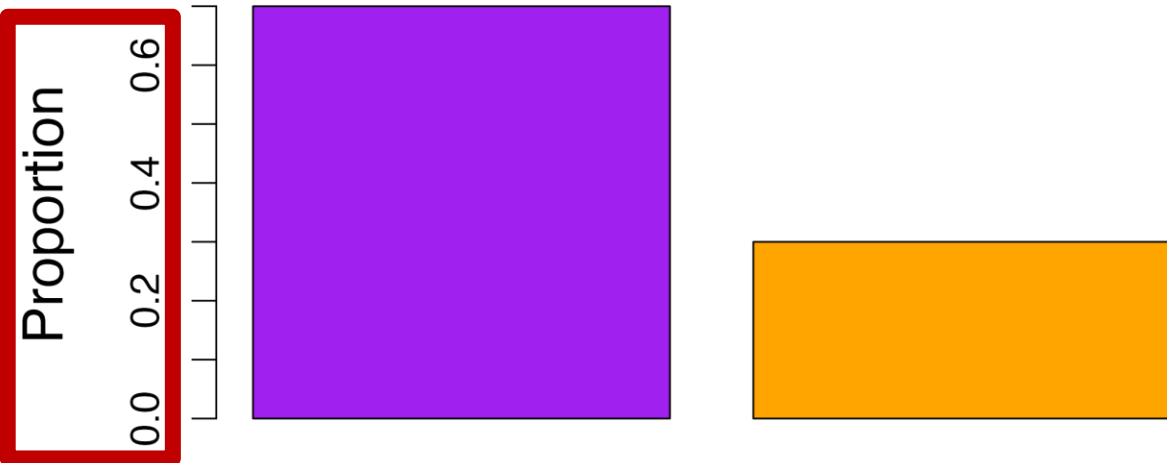
Statistic

$$\hat{p} = 0.74$$

Population and data distribution

Spot the 3 differences!

Different y-axis

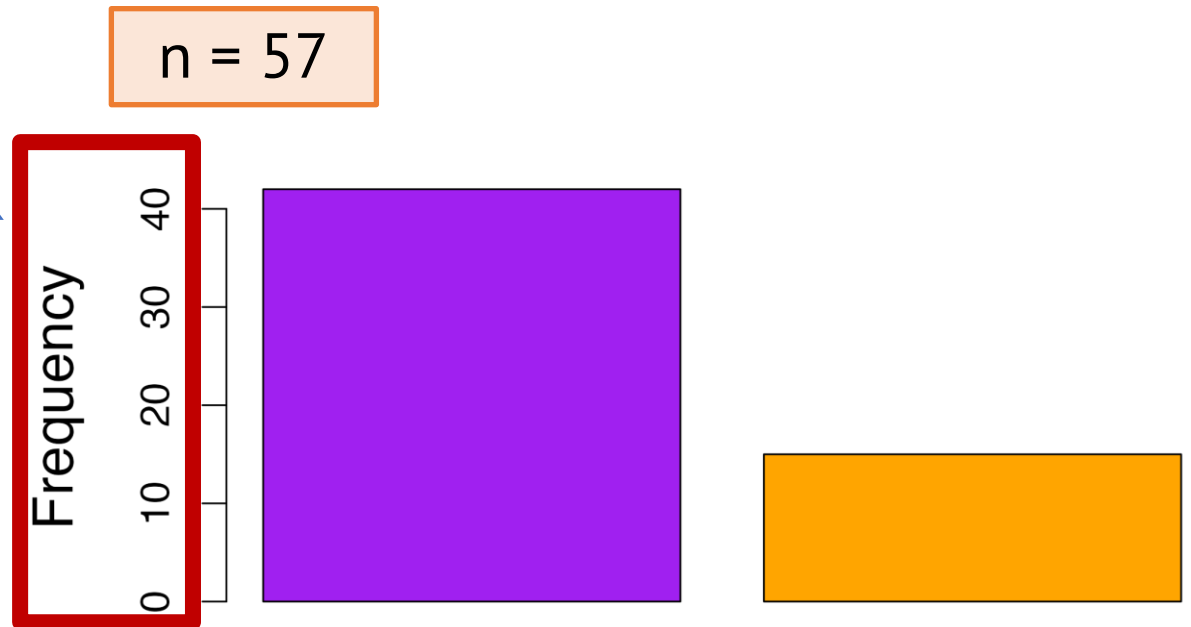


Correct

Incorrect

Parameter

$$p = 0.7$$



Correct

Incorrect

Statistic

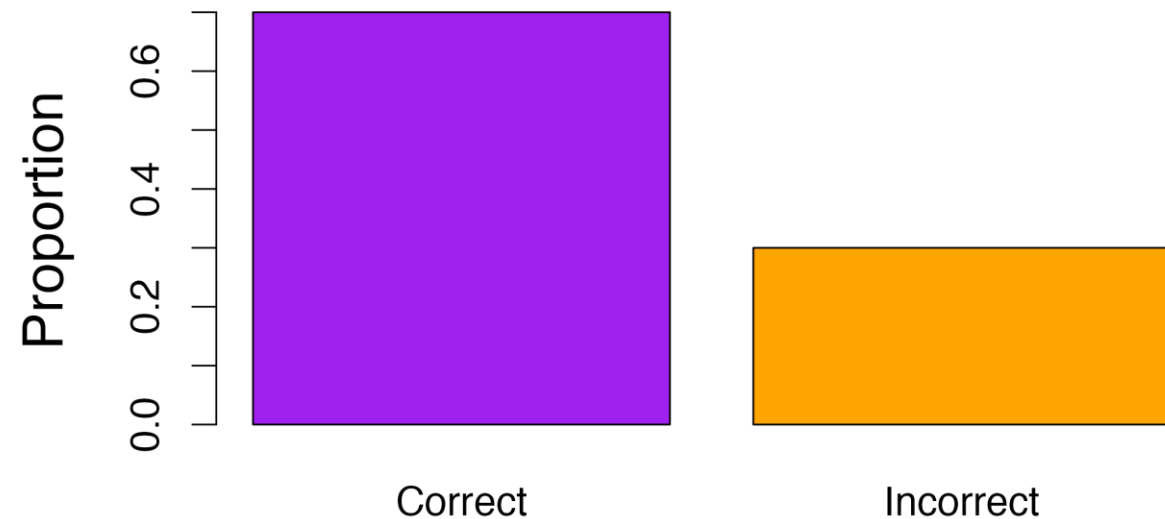
$$\hat{p} = 0.74$$

Population and data distribution

Spot the 3 differences!

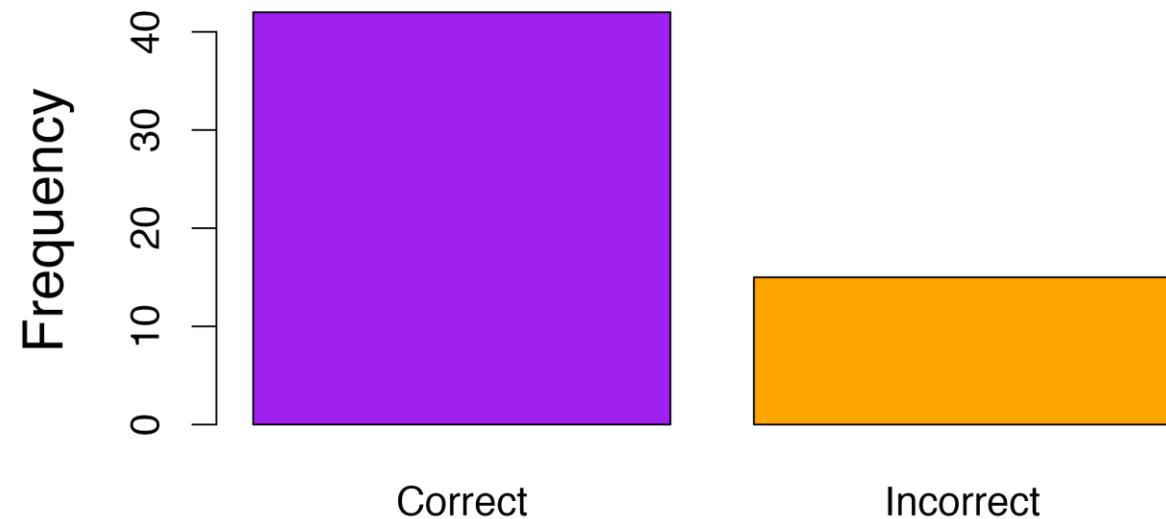
Sample size

$n = 57$



Parameter

$p = 0.7$



Statistic

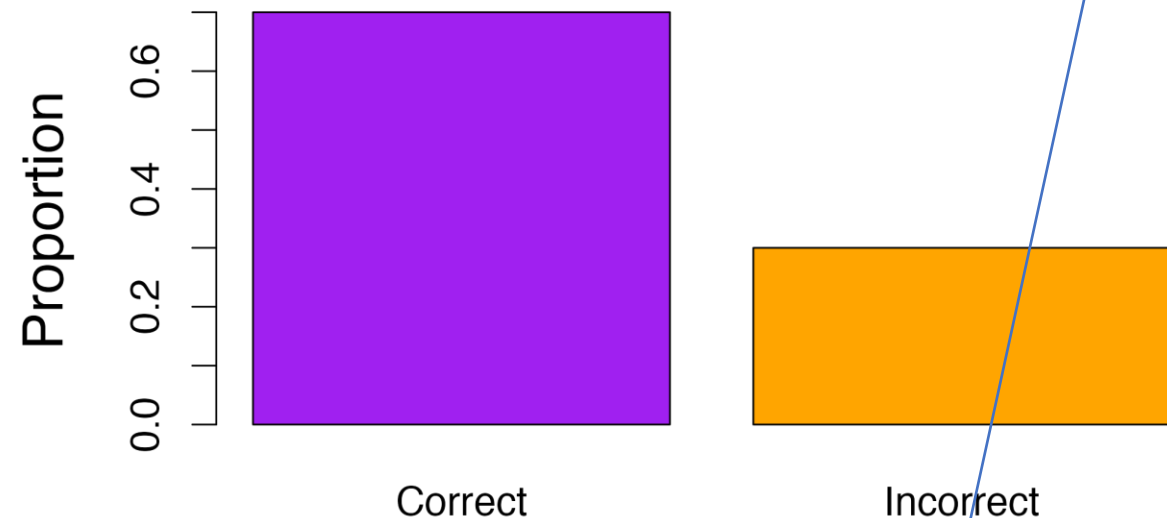
$\hat{p} = 0.74$

Population and data distribution

Spot the 3 differences!

Different notation

$n = 57$

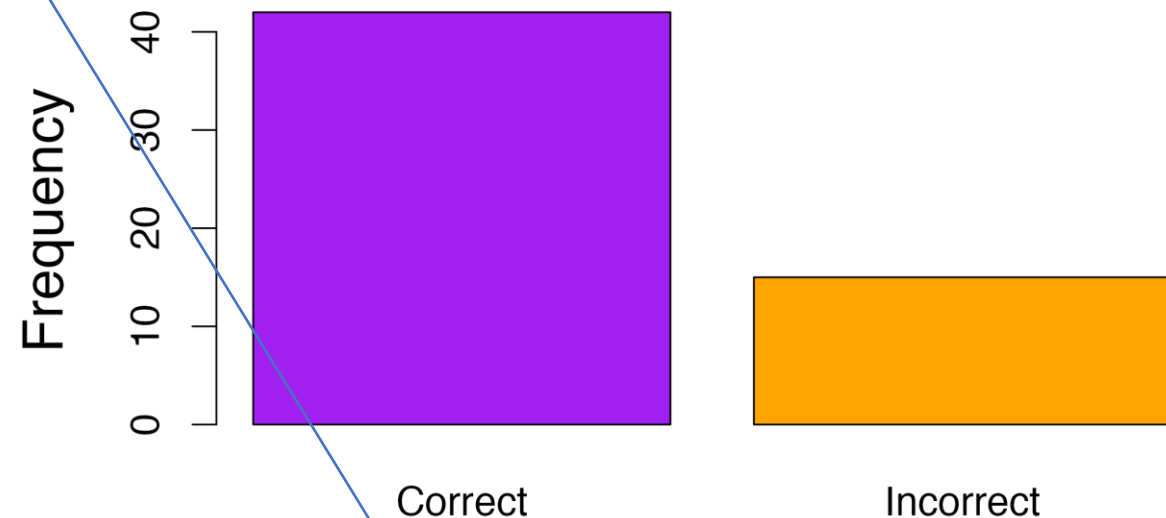


Correct

Incorrect

Parameter

$p = 0.7$



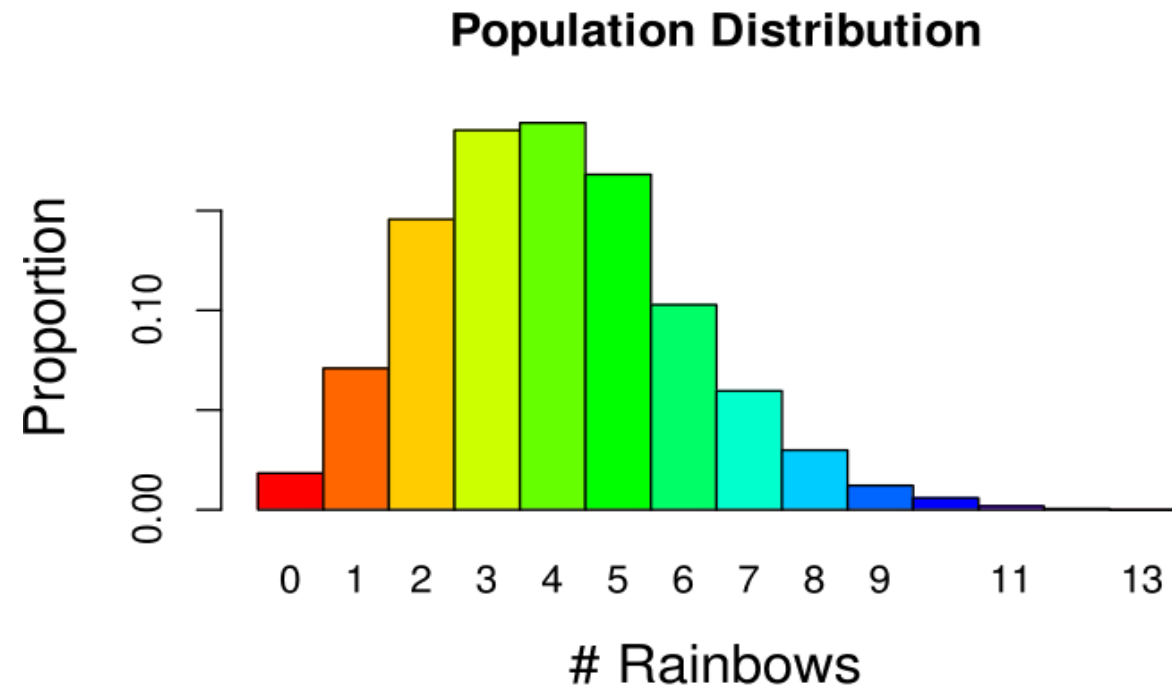
Correct

Incorrect

Statistic

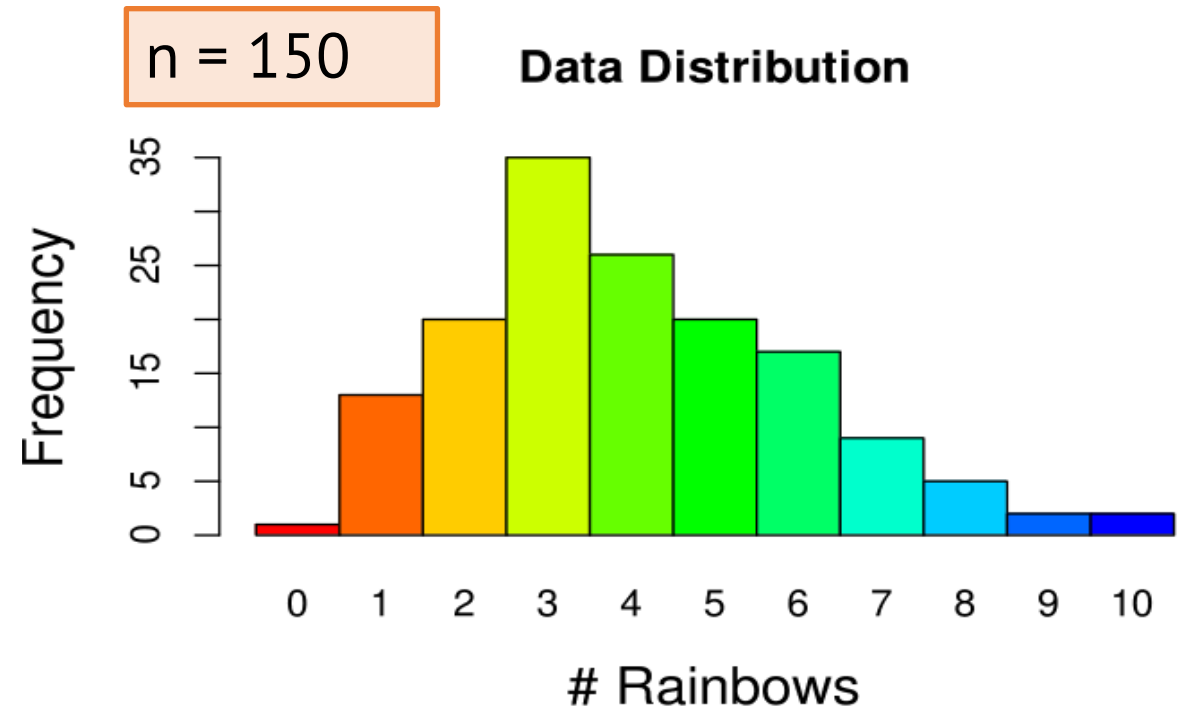
$\hat{p} = 0.74$

Population and data distribution



Parameter

$$\mu = 4$$



Statistic

$$\bar{x} = 3.89$$

If we would **not** have known the
population parameters, how
certain would we be about these
statistics?

Sampling distribution

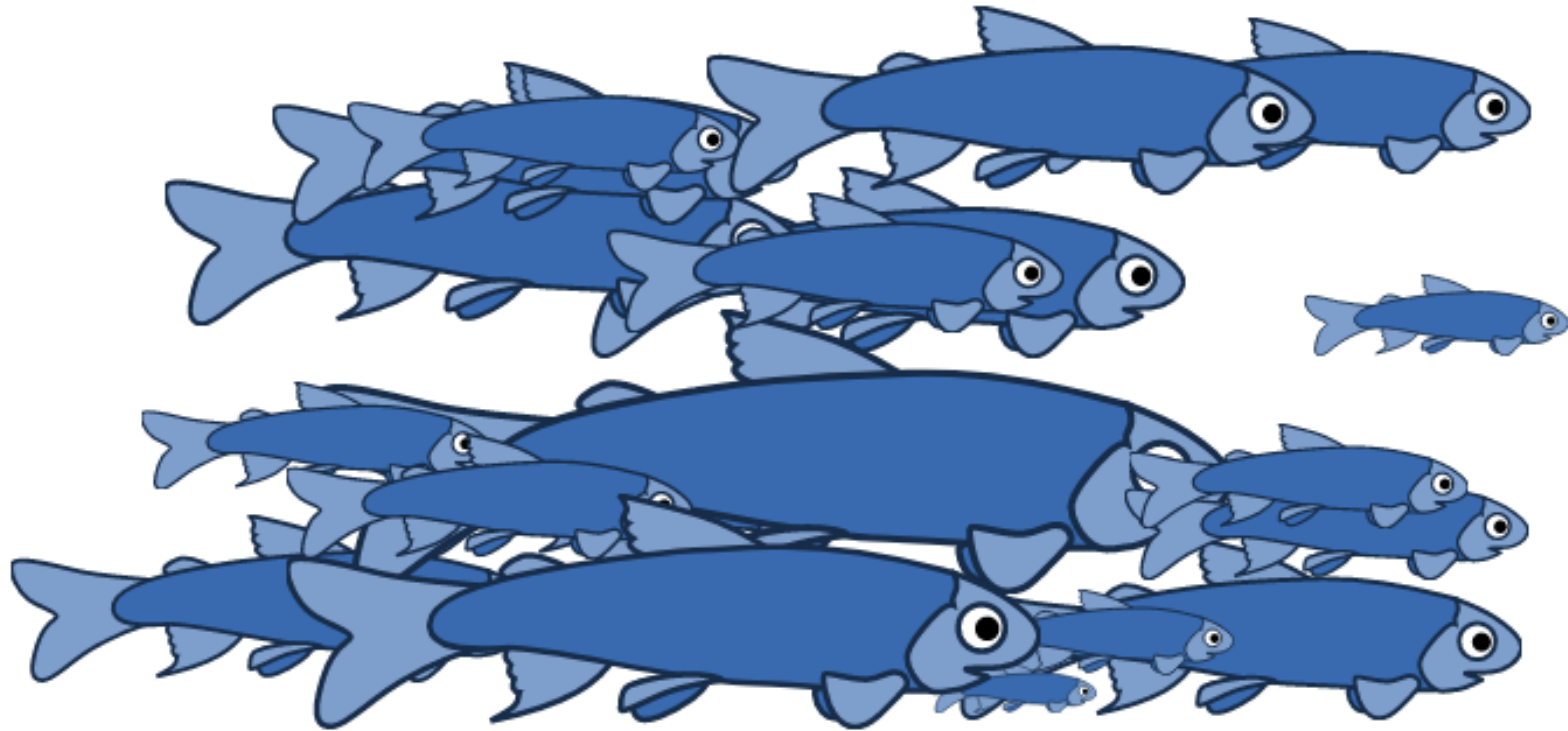
- Every statistic (e.g., proportion, mean, median, IQR, correlation, standard deviation, etc.) has a sampling distribution
- Here we focus on
 - Proportion (relevant for binary categorical variables)
 - Mean (relevant for quantitative variables)

Sampling distribution: A sampling distribution of a statistic is a probability distribution that specifies probabilities for the possible values the statistic can take (Agresti, p. 356).

Sampling distribution

- Every statistic (e.g., proportion, mean, median, IQR, correlation, standard deviation, etc.) has a sampling distribution
- Here we focus on
 - Proportion (relevant for binary categorical variables)
 - Mean (relevant for quantitative variables)

We now start reasoning at the level where we are **imagining** repeating an experiment multiple times and computing a statistic. For instance, we imagine collecting 10,000 samples of size 10. This results in 10,000 means: one for each separate experiment.



<https://www.zoology.ubc.ca/~whitlock/Kingfisher/SamplingNormal.htm>

Three distributions (Agresti, p. 367)

Population distribution: The distribution of the variable of interest in the population from which we sample.

Data distribution: The distribution of the variable of interest computed from the sample and the distribution we see in practice.

Sampling distribution: The distribution of the sample statistic (e.g., sample proportion or sample mean) we would get when repeatedly drawing samples of size n from the population

The sampling distribution is a bit like a unicorn: we can imagine what one looks like, we can appreciate its beauty, and we can wonder at its magical feats, but the sad truth is that you'll never see a real one. They both exist as ideas rather than physical things. You would never go out and actually collect thousands of samples and draw a frequency distribution of their means, instead very clever statisticians have used maths to describe what these distributions look like and how they behave. Likewise, you'd be ill-advised to search for unicorns.

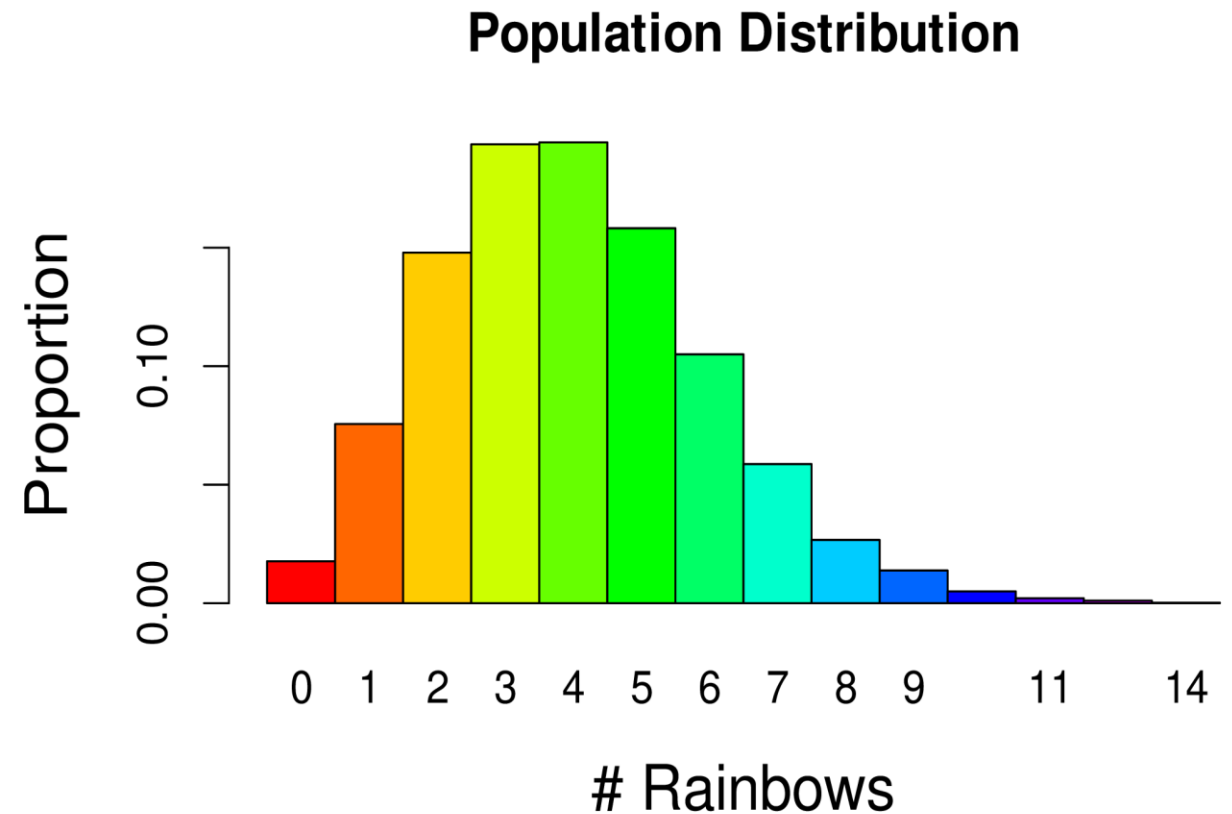
- Andy Field (Discovering Statistics Using JASP, pp. 73–74)

Overview of Today

1. Population distribution vs data distribution
2. Sampling distribution
 1. **Distribution of a quantitative variable**
 2. Distribution of a binary categorical variable
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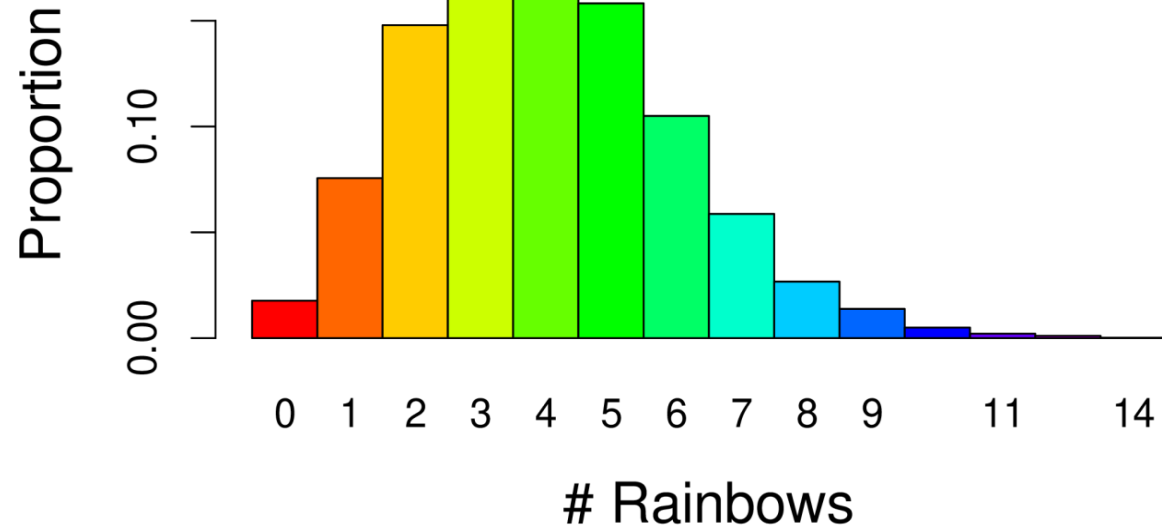
Case 2: Rainbows

- How many rainbows have you seen this year?



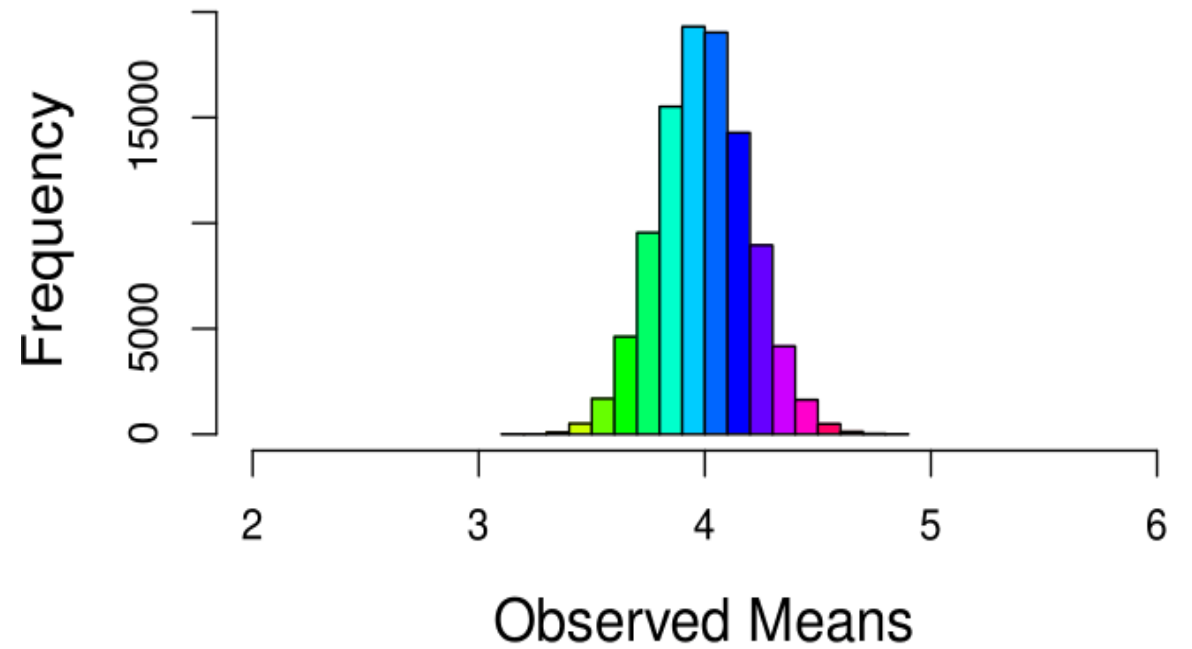
What is the shape of the sampling distribution of means?

Population Distribution



$$\mu = 4, \sigma = 2$$

Sampling distribution mean
After 100,000 samples (n=100)



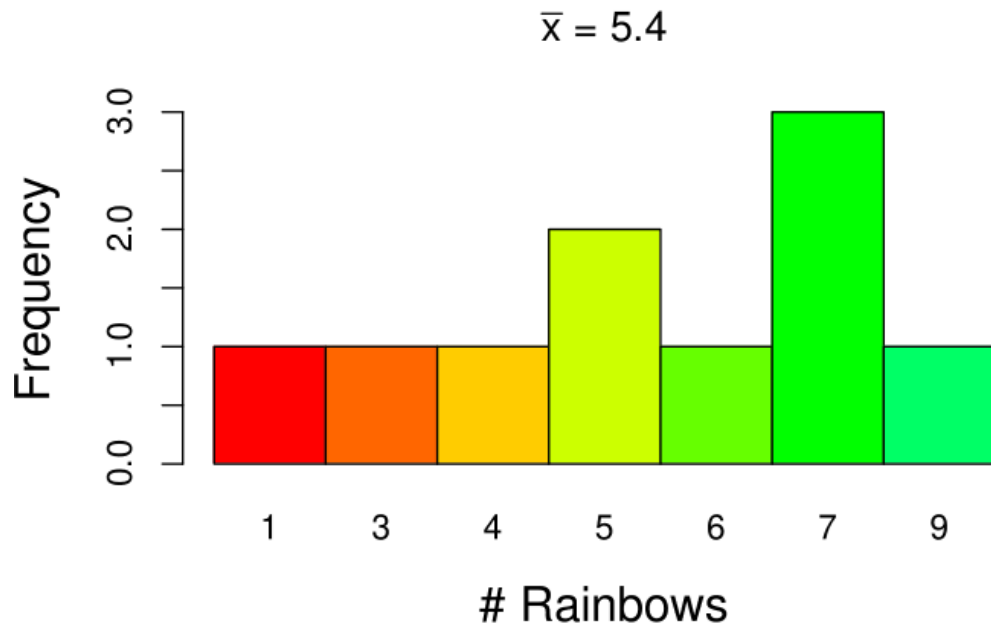
What is the shape of the sampling distribution of means?

- It turns out, that this is usually a normal distribution with
 - Mean: μ
 - Standard deviation: $\sigma / \sqrt{n}$
- For now, let's assume we know the parameters in our example:
 - $\mu = 4$
 - $\sigma = 2$

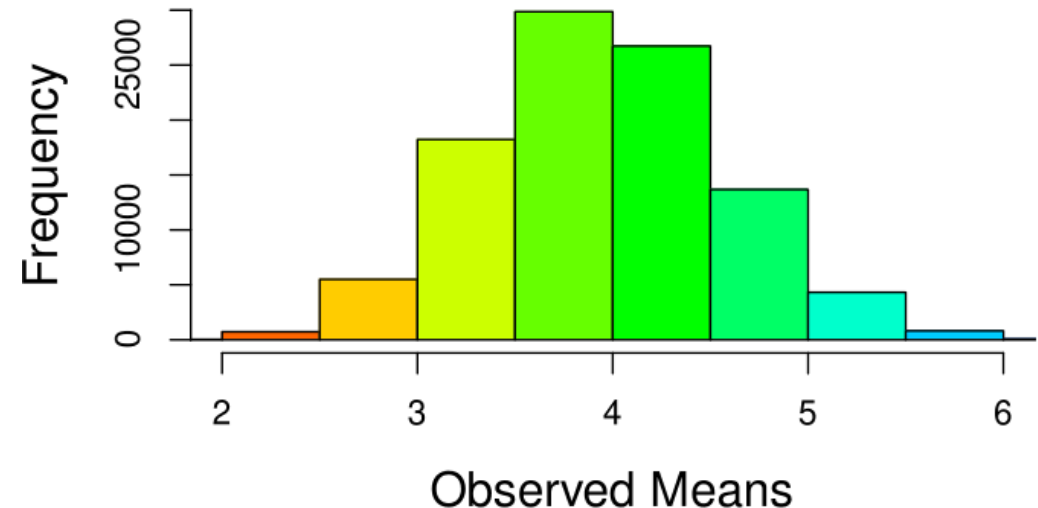
Next week, we see how parameter estimation works in practice, when we do **not** know the parameter values!

What is the shape of the sampling distribution of means?

One observed sample and its observed mean ($n=10$)



Sampling distribution mean
After 100,000 samples ($n=10$)

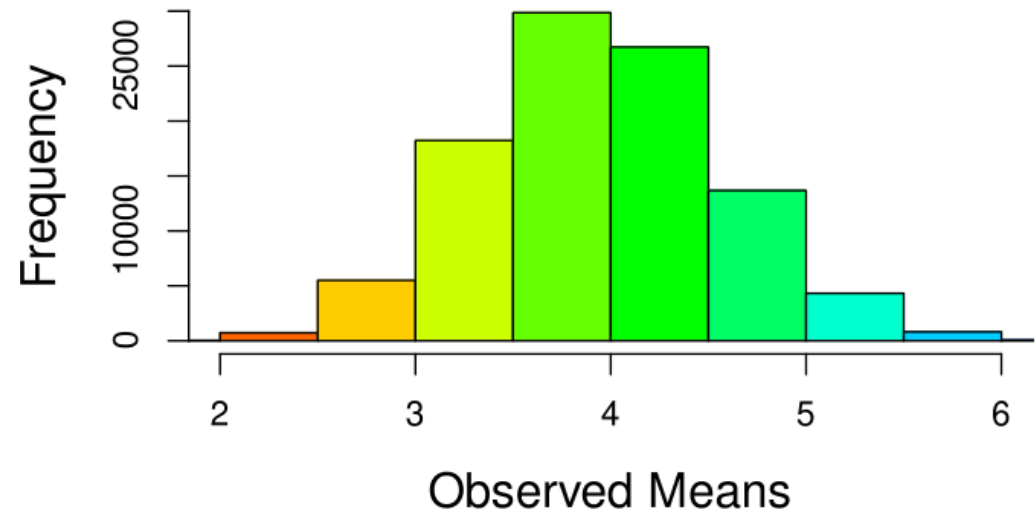


What is the shape of the sampling distribution of means?

$$\text{mean} = \mu = 4$$

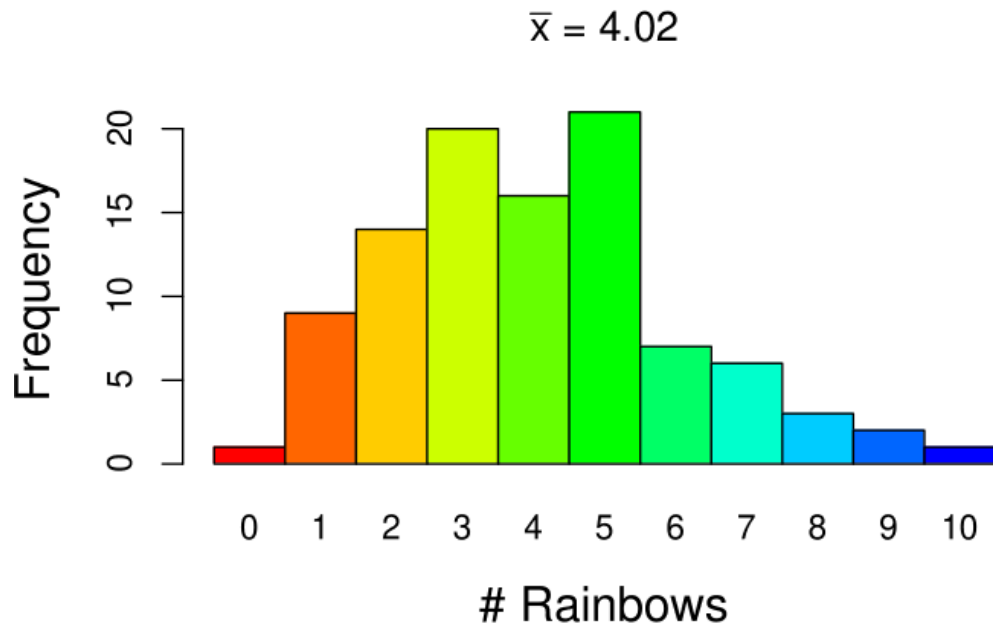
$$sd = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{10}} = 0.63$$

Sampling distribution mean
After 100,000 samples ($n=10$)

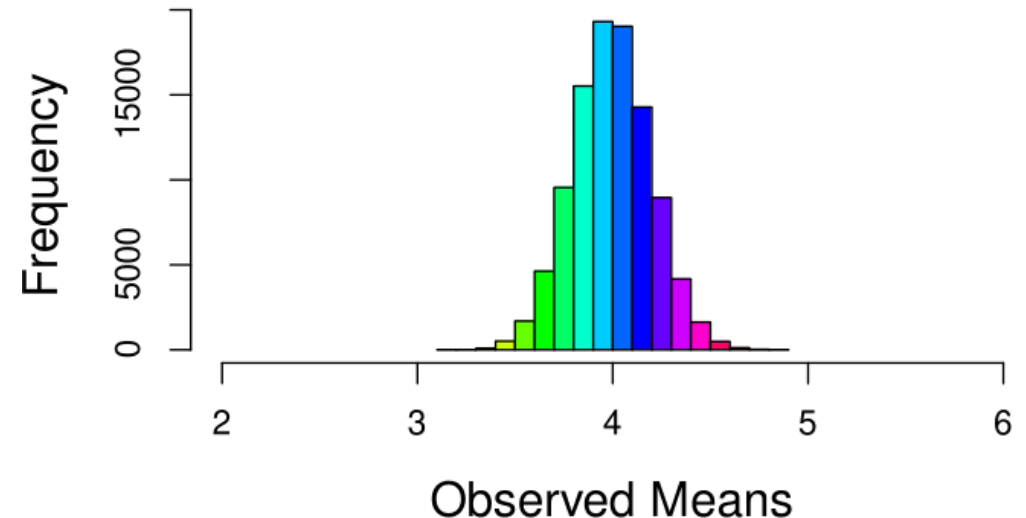


What is the shape of the sampling distribution of means?

One observed sample and its observed mean ($n=100$)



Sampling distribution mean
After 100,000 samples ($n=100$)

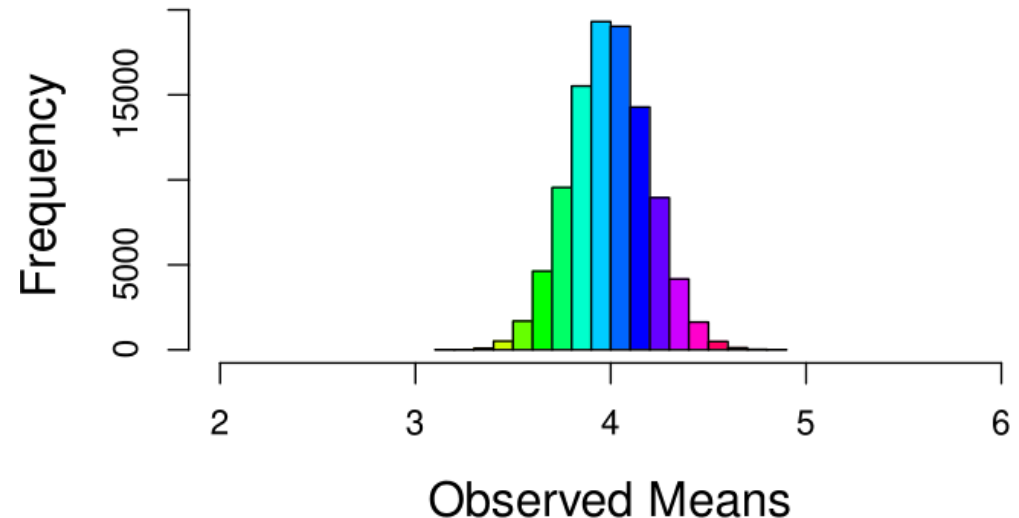


What is the shape of the sampling distribution of means?

$$\text{mean} = \mu = 4$$

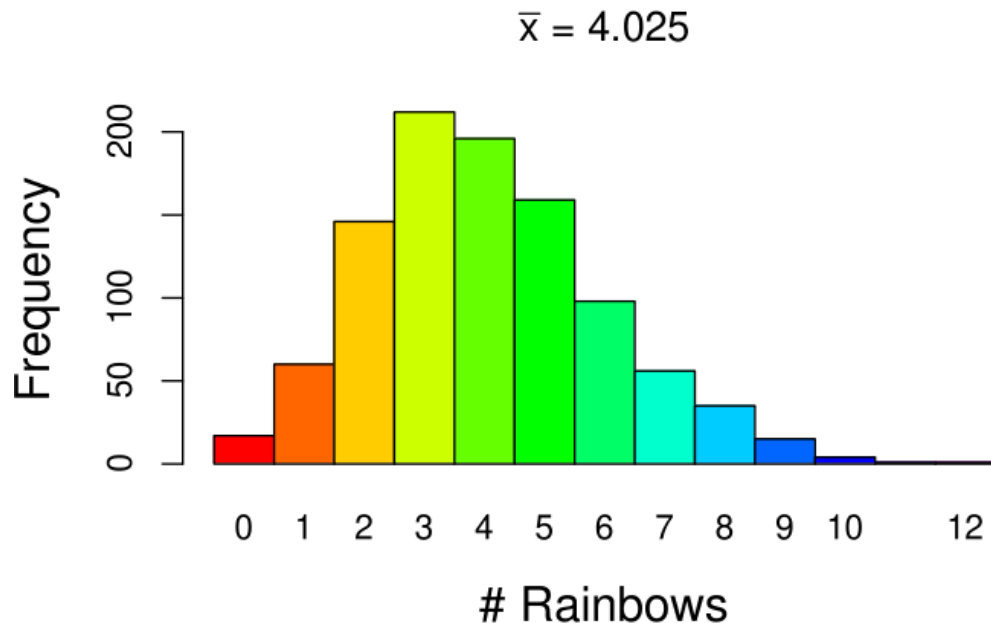
$$sd = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{100}} = 0.2$$

Sampling distribution mean
After 100,000 samples ($n=100$)

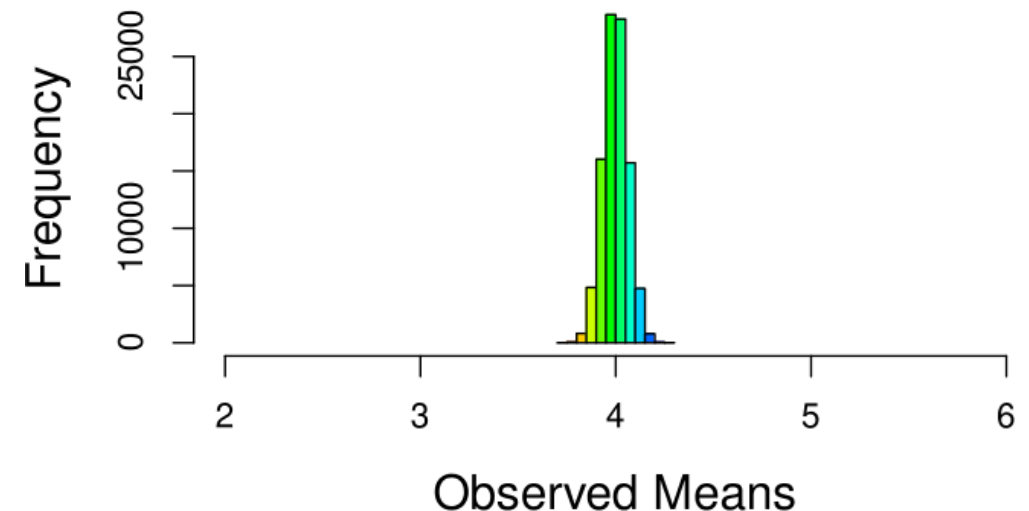


What is the shape of the sampling distribution of means?

One observed sample and its observed mean ($n=1,000$)



Sampling distribution mean
After 100,000 samples ($n=1,000$)

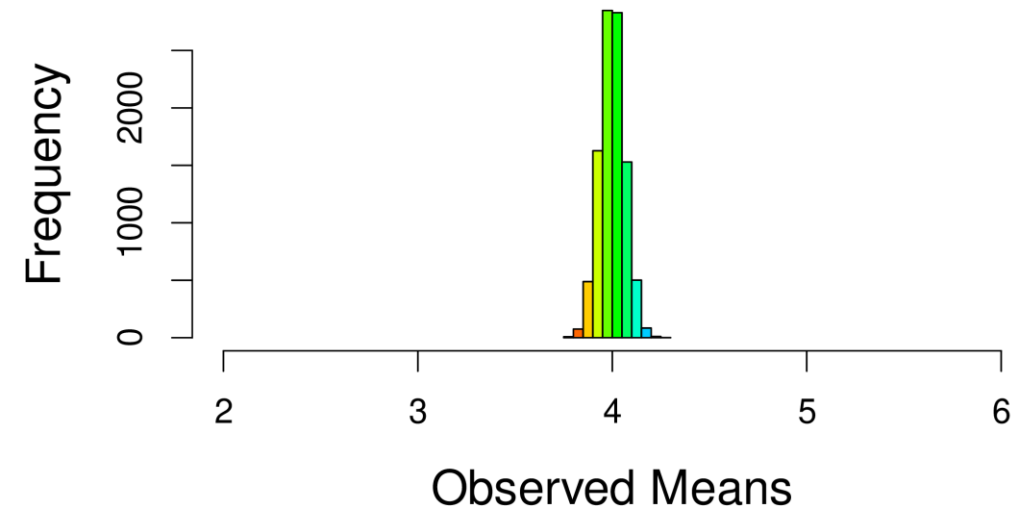


What is the shape of the sampling distribution of means?

$$\text{mean} = \mu = 4$$

$$sd = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{1000}} = 0.063$$

Sampling distribution mean
After 100,000 samples ($n=1,000$)



Overview of Today

1. Population distribution vs data distribution
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 - 2. Distribution of a binary categorical variable**
3. Central Limit Theorem
4. Excel & Demo
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 - Example exam questions

What is the shape of the sampling distribution of proportions?

- We are drawing repeatedly and independently from a population that has two categories (e.g., correct/incorrect in the beer tasting)
- The probability of one category (correct) is $P(\text{correct}) = p = 0.7$
 - The population proportion p
 - $P(\text{incorrect}) = 1 - p = 1 - 0.7 = 0.3$
- For every draw, p is the same
- What is the distribution of the proportion of people that are correct, after n draws?

⑦ Binomial distribution (of the number correct)

What is the shape of the sampling distribution of proportions?

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- What is the distribution of the proportion of people that are correct after n draws?

⑦ Binomial distribution (of the number correct)

Note: this is only true if we draw *with replacement*.

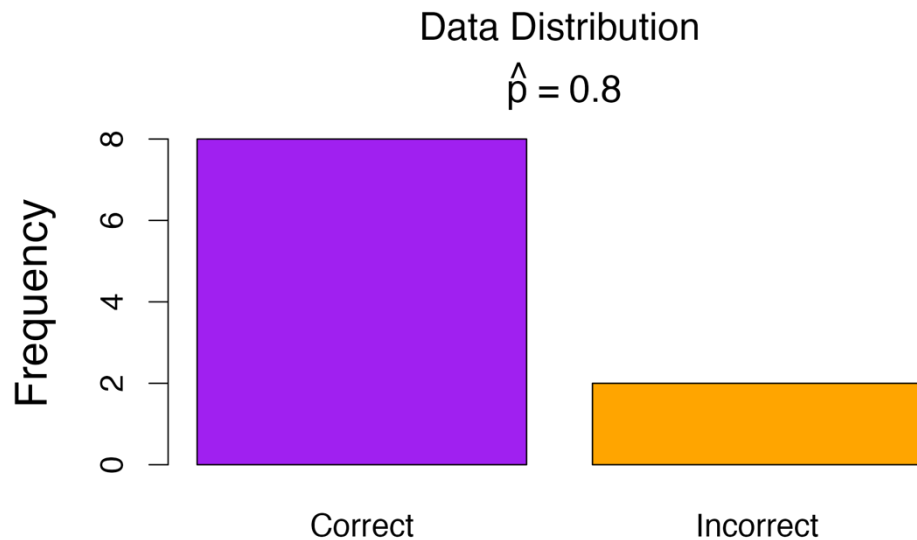
In practice, for large populations this does not make a difference.

What is the shape of the sampling distribution of proportions?

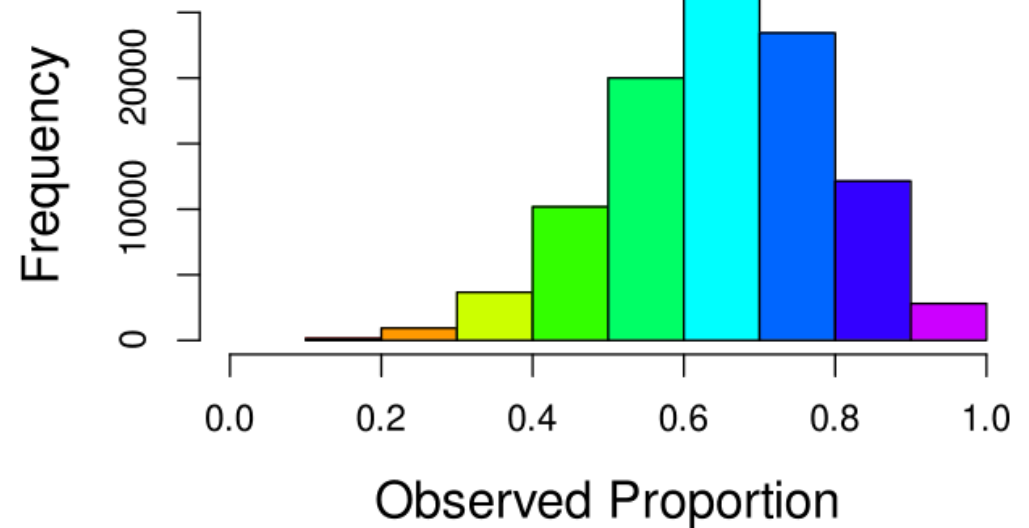
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- ⑦ Binomial distribution (of the number correct)**

What is the shape of the sampling distribution of proportions?

One observed sample and its observed proportion (n=10)

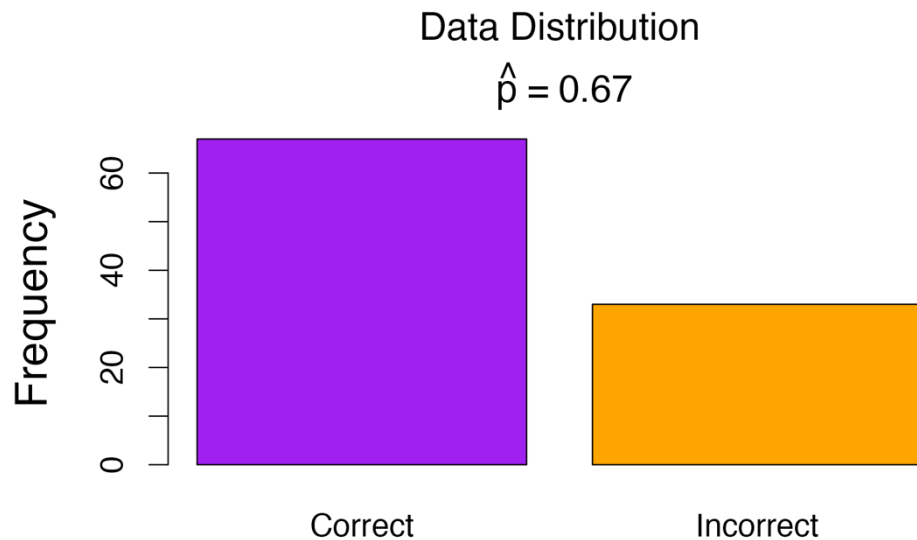


Sampling distribution proportion
After 100,000 samples (n=10)

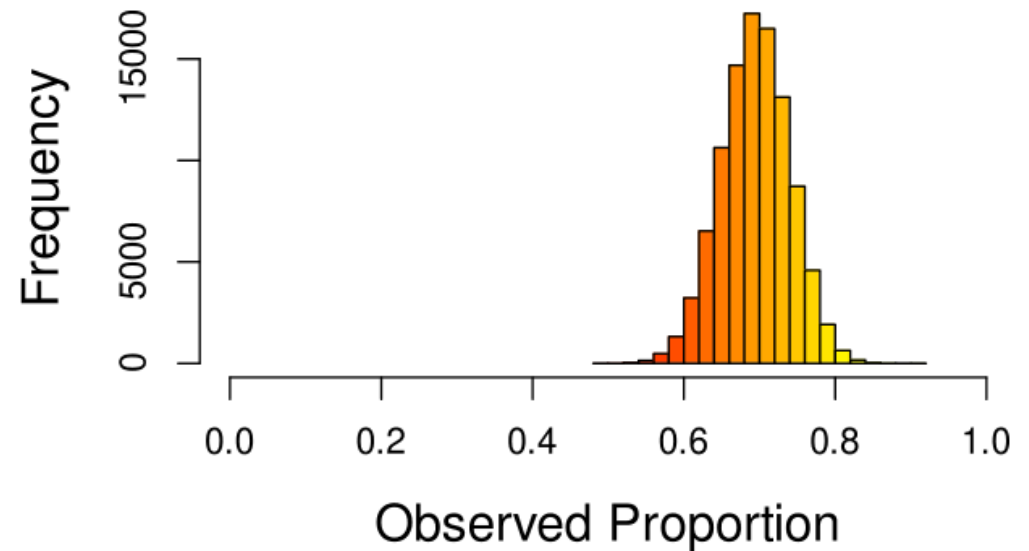


What is the shape of the sampling distribution of proportions?

One observed sample and its observed proportion ($n=100$)

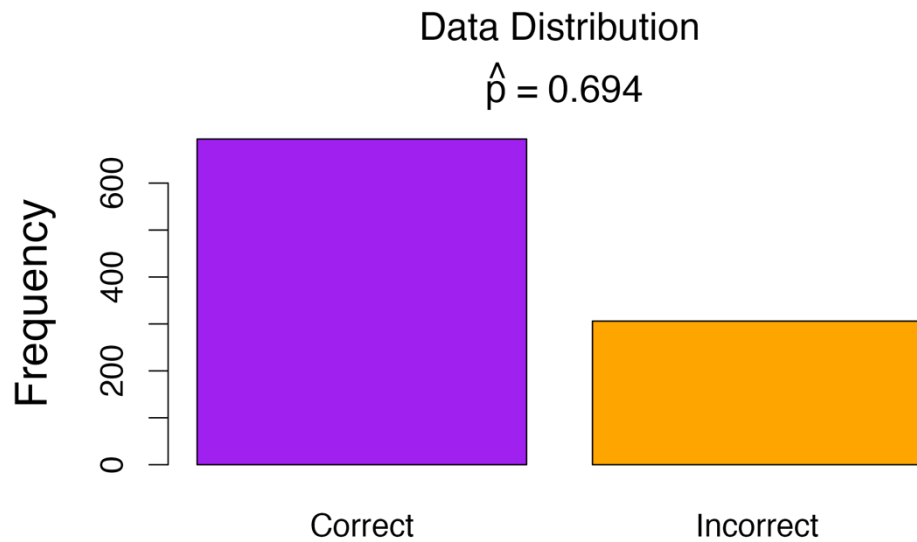


Sampling distribution proportion
After 100,000 samples ($n=100$)

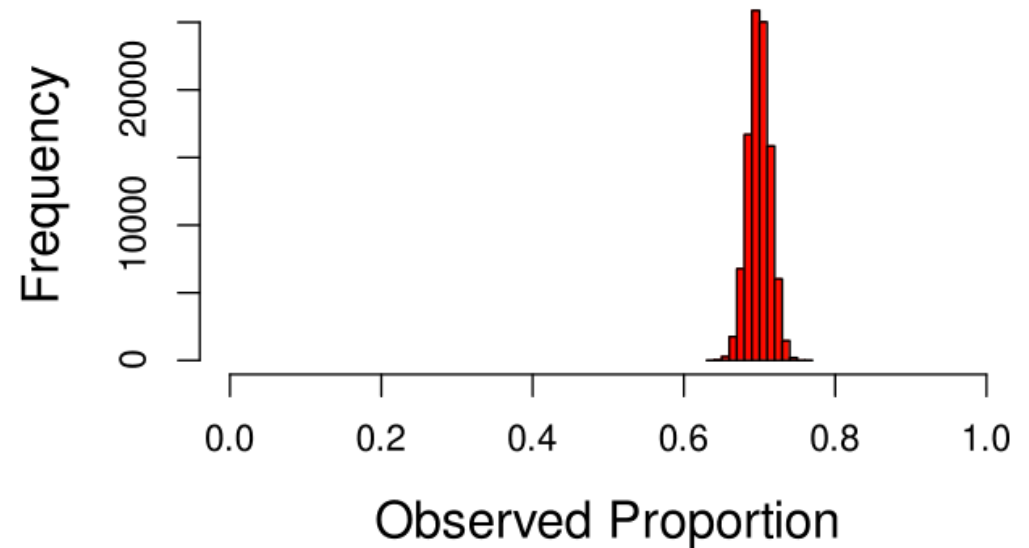


What is the shape of the sampling distribution of proportions?

One observed sample and its observed proportion (n=1,000)



Sampling distribution proportion
After 100,000 samples (n=1,000)



What is the shape of the sampling distribution of proportions?

The sampling distribution of the proportion $\hat{p}$:

mean: p

standard deviation: $\sqrt{\frac{p(1-p)}{n}}$

- The mean of the sampling distribution is the population parameter
- The standard deviation indicates the variability of the sampling distribution: How far can a statistic deviate from the parameter?
- A larger sample (n) means that the standard deviation of the sampling distribution **decreases**

What is the shape of the sampling distribution of proportions?

The sampling distribution of the proportion $\hat{p}$:

mean: p

standard deviation: $\sqrt{\frac{p(1-p)}{n}}$

Count or proportion? Same data, same shape, same uncertainty

	Number correct (X)	Proportion correct ($\hat{p} = X/n$)
Distribution	binomial (Lecture 9)	binomial, divided by n
Mean	np	p
SD	$\sqrt{np(1-p)}$	$\sqrt{\frac{p(1-p)}{n}}$

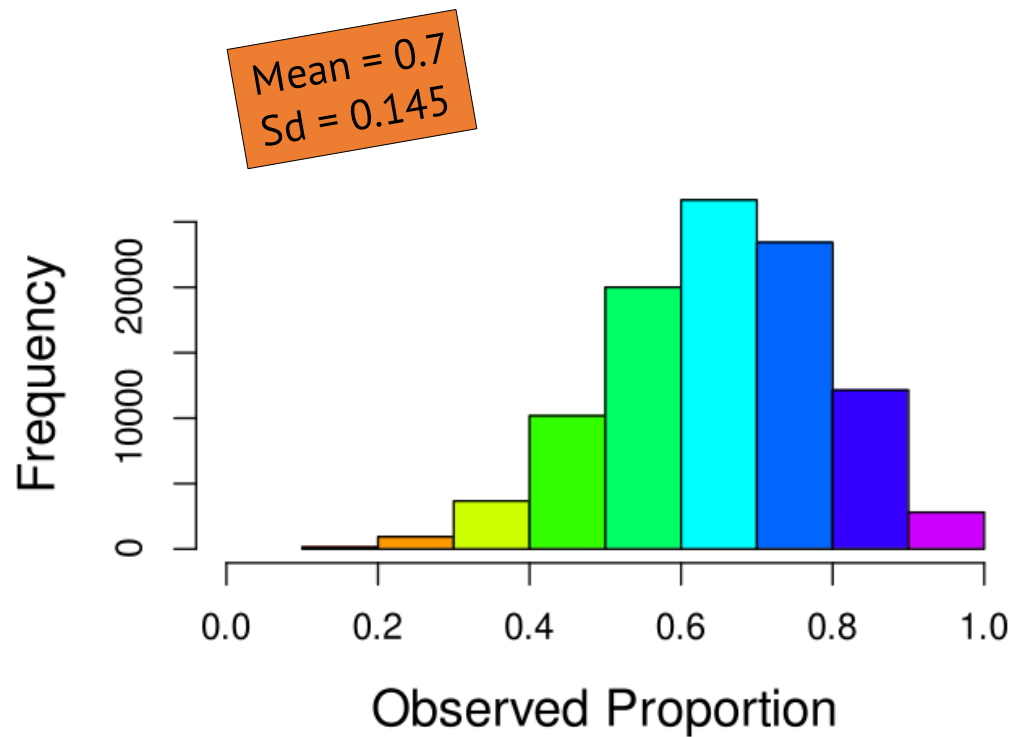
7 out of 10 correct = 0.7: divide every value by n , only the axis changes.

mean = $p = 0.7$

Standard deviation = $\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{p(1-p)}{10}} = 0.145$



Sampling distribution proportion
After 100,000 samples ($n=10$)



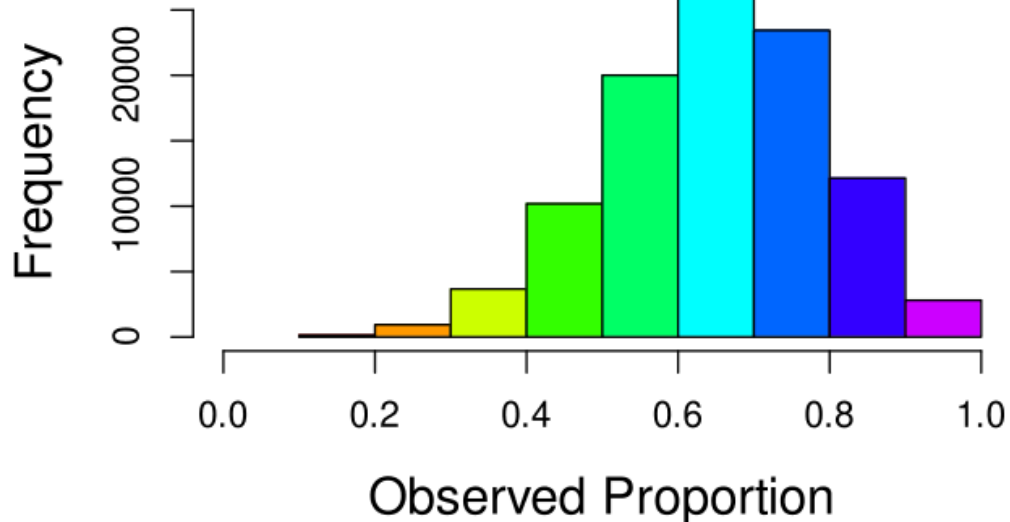
mean = $p = 0.7$

$$\text{Standard deviation} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{p(1-p)}{10}} = 0.145$$



Sampling distribution proportion
After 100,000 samples ($n=10$)

Mean = 0.7
Sd = 0.145



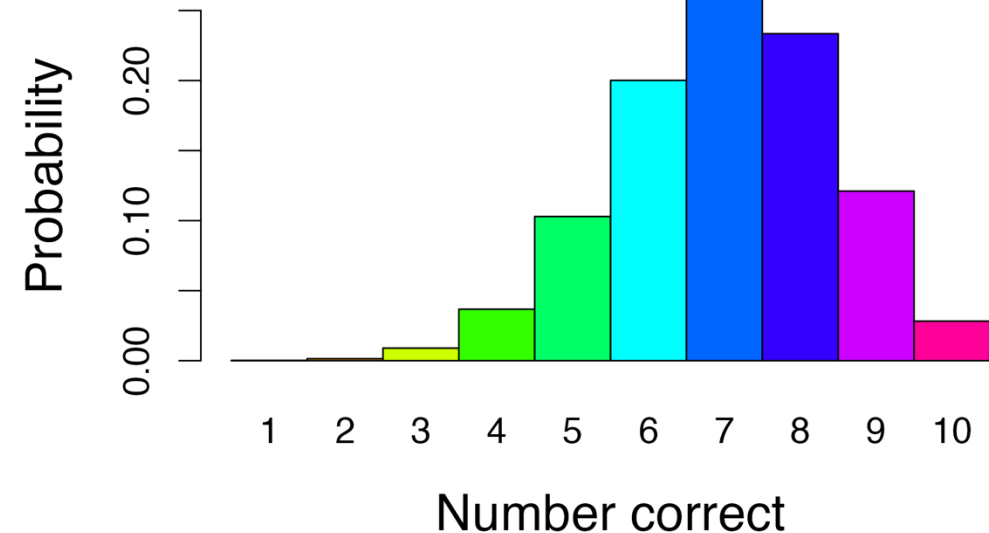
mean = $np = 7$

$$\text{Standard deviation} = \sqrt{np(1-p)} = 1.45$$



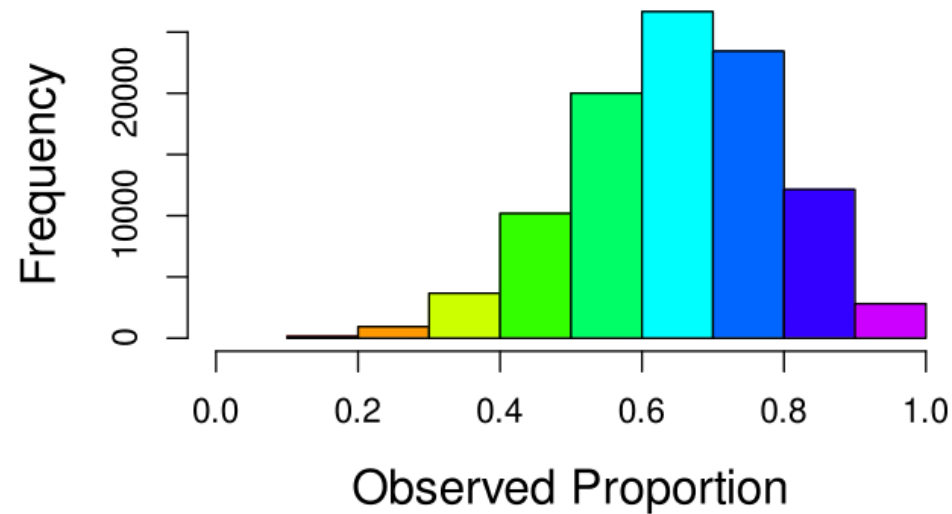
Number correct (binomial)
 $n=10, p=0.7$

Mean = 7
Sd = 1.45



What is the shape of the sampling distribution of proportions?

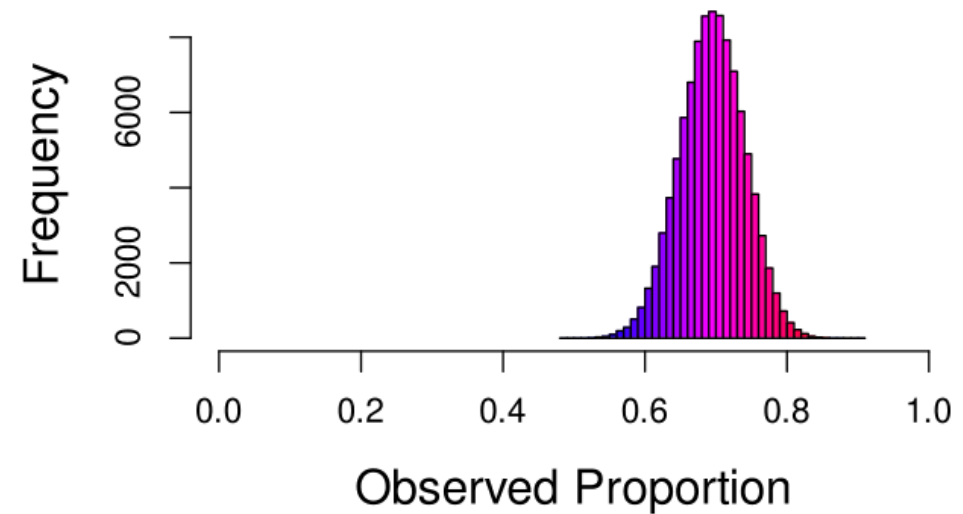
Sampling distribution proportion
After 100,000 samples ($n=10$)



mean = $p = 0.7$

$$\text{Standard deviation} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{p(1-p)}{10}} = 0.145$$

Sampling distribution proportion
After 100,000 samples ($n=100$)



mean = $p = 0.7$

$$\text{Standard deviation} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{p(1-p)}{100}} = 0.046$$

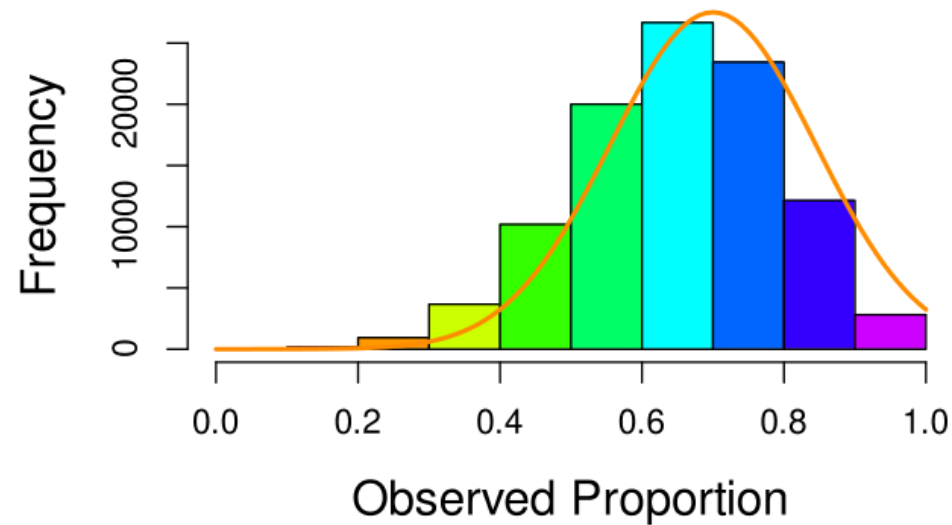
What is distribution

Approximately normally distributed when:

$$np \geq 15$$

$$n(1-p) \geq 15$$

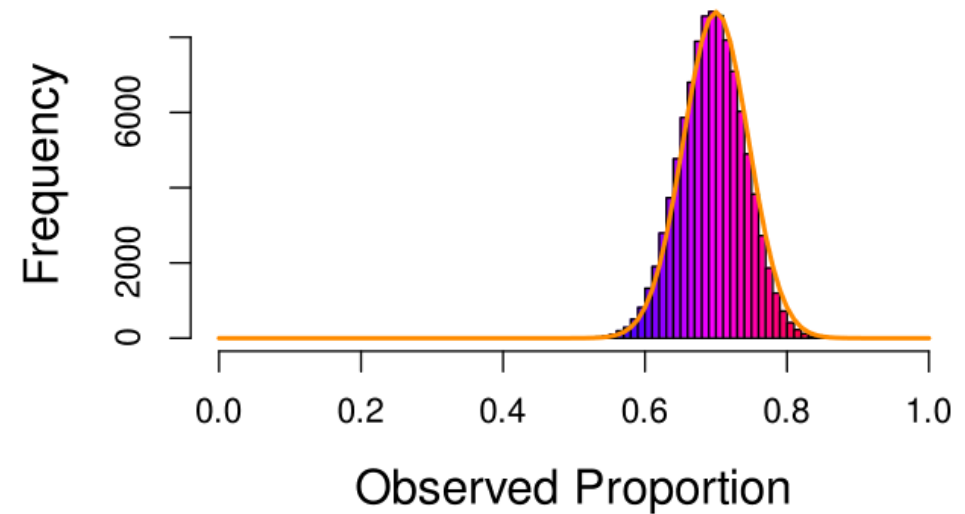
Sampling distribution proportion
After 100,000 samples ($n=10$)



$$\text{mean} = p = 0.7$$

$$\text{Standard deviation} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{p(1-p)}{10}} = 0.145$$

Sampling distribution proportion
After 100,000 samples ($n=100$)

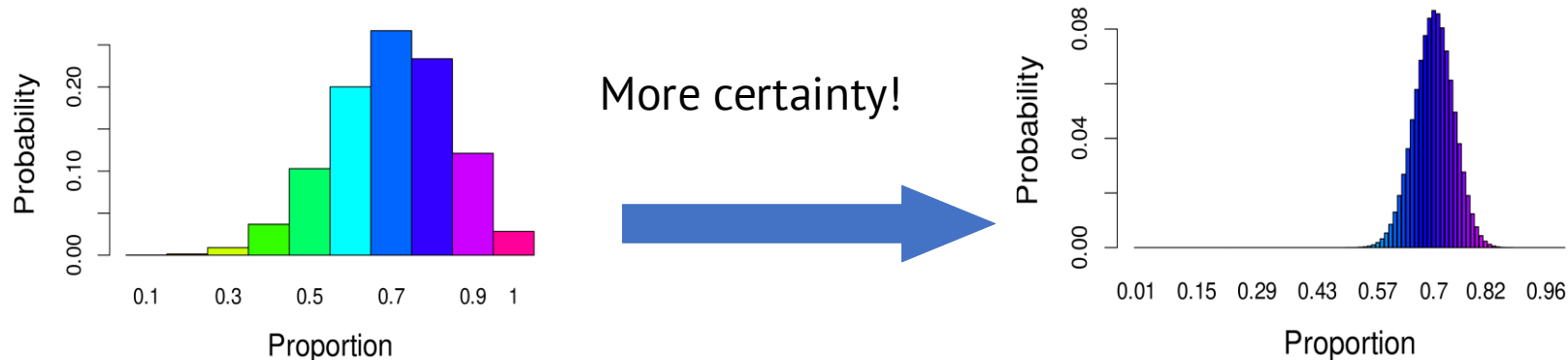


$$\text{mean} = p = 0.7$$

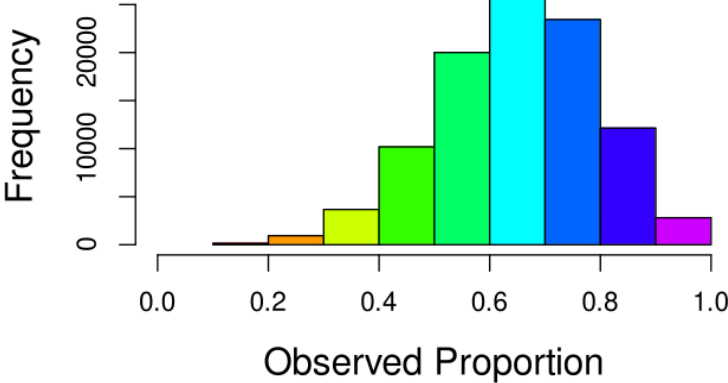
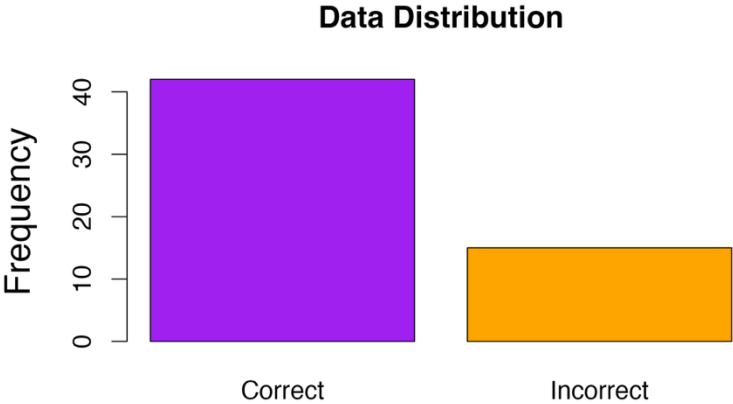
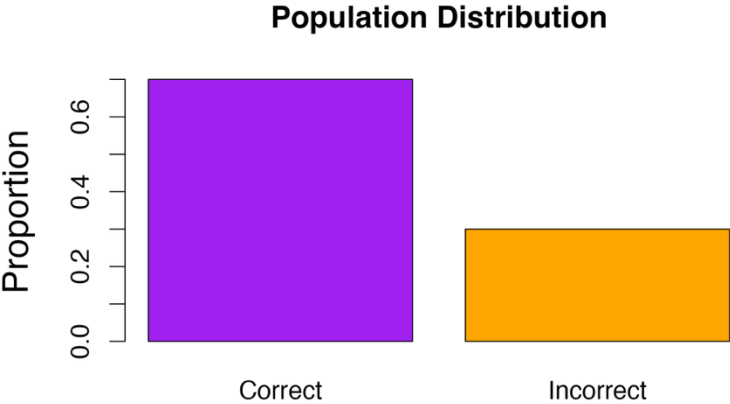
$$\text{Standard deviation} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{p(1-p)}{100}} = 0.046$$

So, what have we learned now?

- We can now estimate how likely it is that the statistics we get from our experiment or observational study are good representations of the population
- We know how likely it is that we will get the same statistics if we would have collected another random sample
- We know to what extent we can trust our inference to the population → Validities & proper methodology



Sampling distribution proportion
After 100,000 samples ($n=10$)



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1. Population distribution vs data distribution
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 2. Distribution of a binary categorical variable
- 3. Central Limit Theorem**
4. Excel & Demo
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Central Limit Theorem

Central Limit Theorem: For any large enough sample, the sampling distribution of the mean is approximately normally distributed with

mean: μ

Standard deviation: $\sigma/\sqrt{n}$

- This is *irrespective* of the shape of the distribution
- <http://www.zoology.ubc.ca/~whitlock/Kingfisher/CLT.htm>

Central Limit Theorem

This is not an all-or-nothing threshold,
but we generally need around 30
observations

Central Limit Theorem: For any large enough sample, the sampling distribution of the mean is approximately normally distributed with

mean: μ
Standard error (SE): $\sigma/\sqrt{n}$

From now, we denote the standard deviation of the sampling distribution with SE: Standard error (from chapter 8, p405)

Central Limit Theorem

- This means that we can study all kinds of variables (with different distributions) because the sampling distributions of their means converge to a normal distribution
- This means that we only need a limited set of statistical tools to draw inferences about our data
 - Parametric statistics

	Quantitative	Categorical
Quantitative	Correlation Regression	t-test ANOVA
Categorical	Logistic regression	Contingency table

What can we do now?

- When we know the sampling distribution (for instance because of the C.L.T., or because we know or assume it), we can speak about the *probability* of observing a certain value for a statistic.
- For instance, the probability that **42 or more** of the 57 tasters ($\hat{p} \geq 0.7368$) pick the alcoholic beer, when the population proportion is $p = 0.7$.

$$P(\hat{p} \geq 0.7368 | p = 0.7)$$

What can we do now?

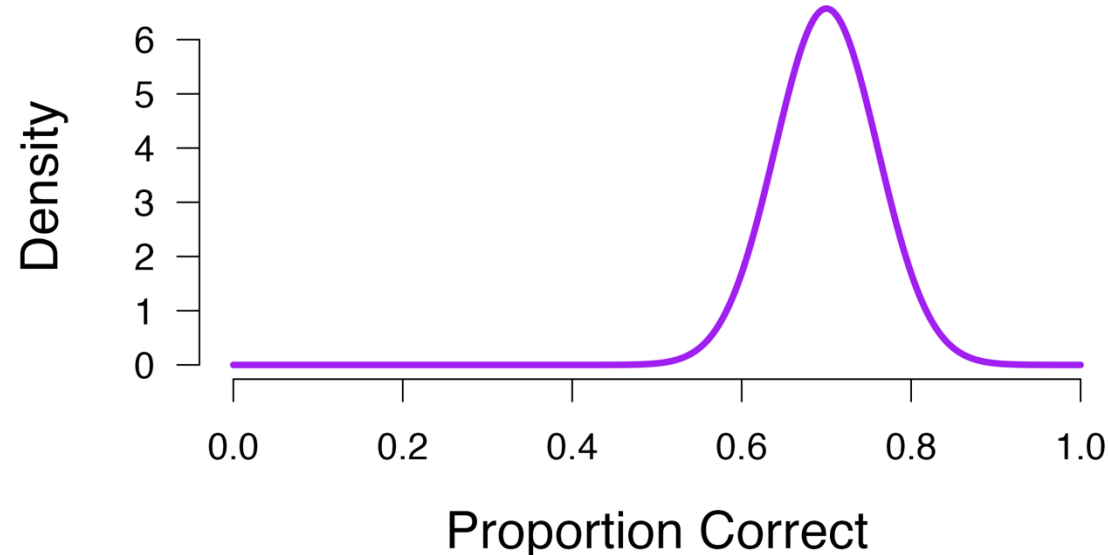
$$P(\hat{p} \geq 0.7368 | p = 0.7)$$

- We know that the proportion is approximately normally distributed:
($np = 39.9 \geq 15$ and $n(1-p) = 17.1 \geq 15$)

$$SE = \sqrt{\frac{0.7(1 - 0.7)}{57}} = 0.0607$$

$p = 0.7$

Sampling distribution



$$P(\hat{p} \geq 0.7368 | p = 0.7)$$

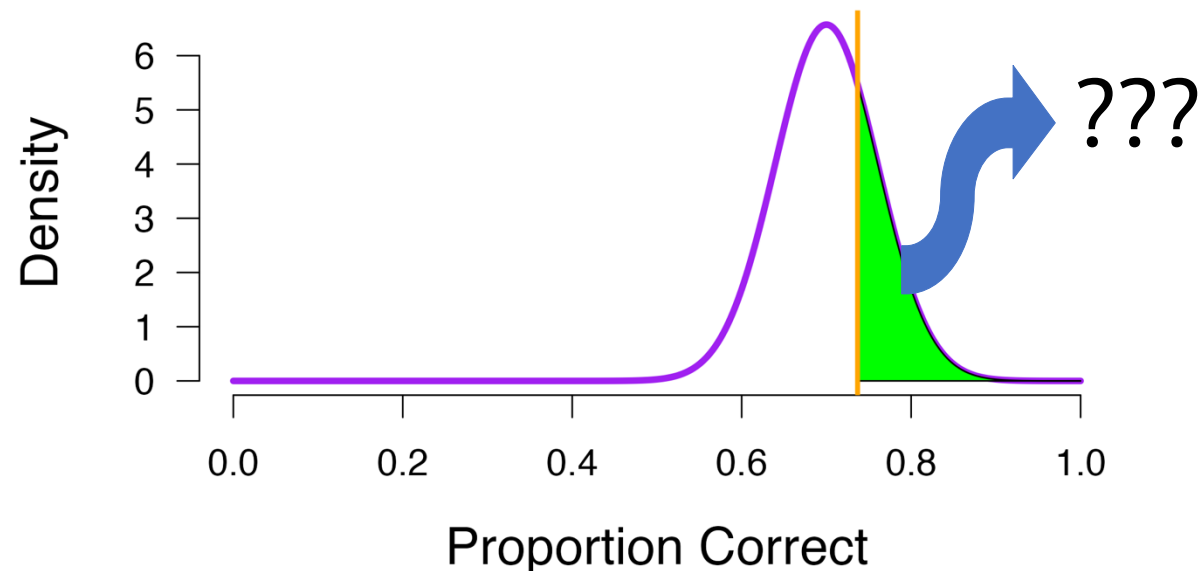
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$p = 0.7$

Sampling distribution



What can we do now?

We are converting the observed proportion to a z-score, so we use the **mean of the sampling distribution (p)**, and the **standard error**

- To know this area, we can transform to z-scores and use Excel

$$SE = 0.0607$$

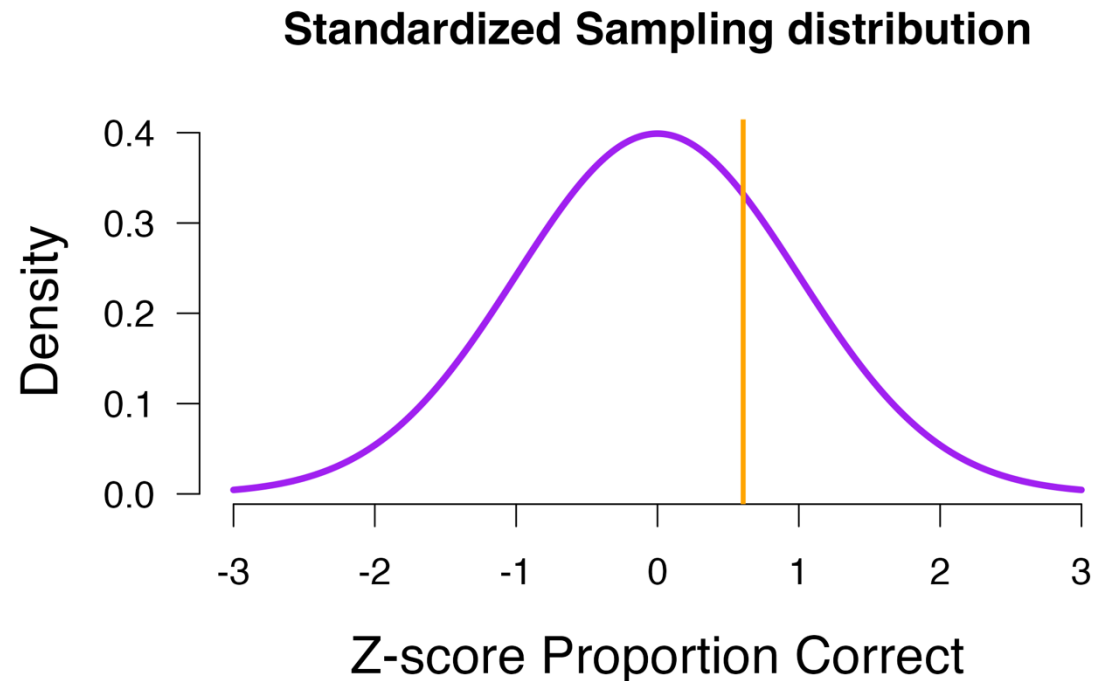
$$z = \frac{\hat{p} - p}{SE} = \frac{0.7368 - 0.7}{0.0607} = 0.6063$$

$$p = 0.7$$

$$P(\hat{p} \geq 0.7368 | p = 0.7) = P(z \geq 0.6063)$$

$$z = \frac{0.7368 - 0.7}{0.0607} = 0.6063$$

- To know this area, we can transform to z-scores and then use Excel



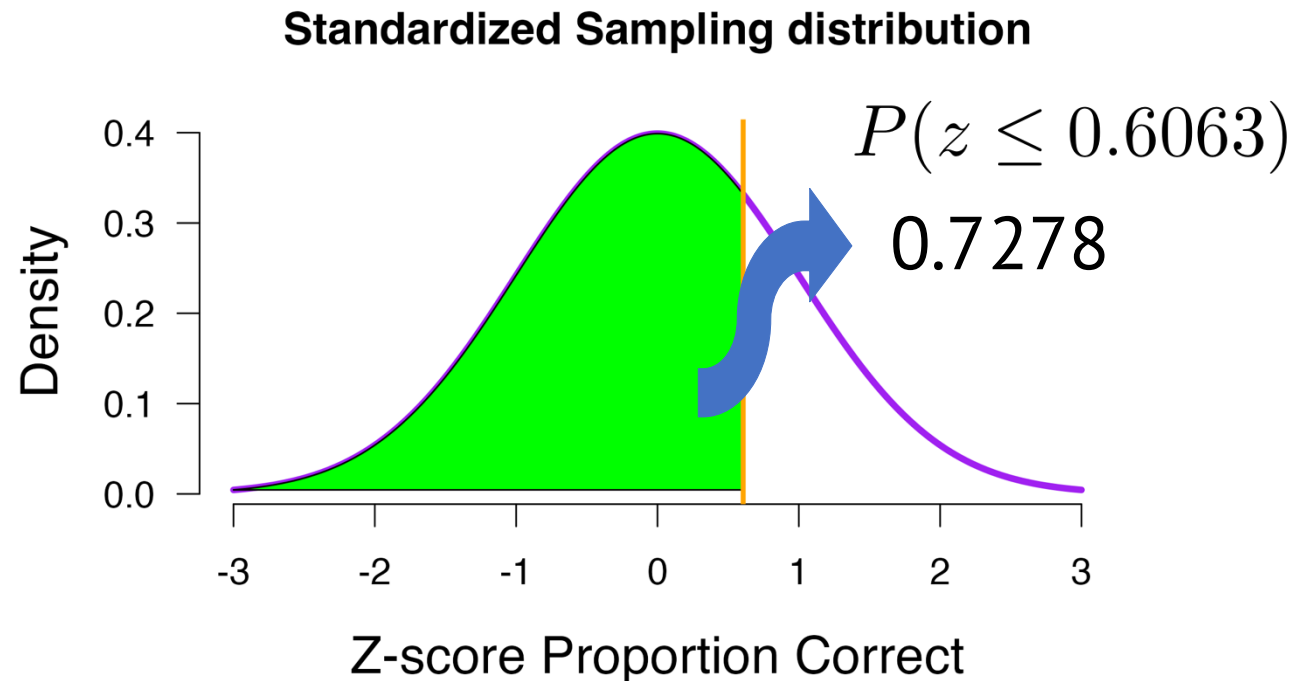
- To know this area, we can use Excel!

- = NORM.DIST(0,6063; 0; 1; TRUE) = 0.7278

$$z = \frac{0.7368 - 0.7}{0.0607} = 0.6063$$

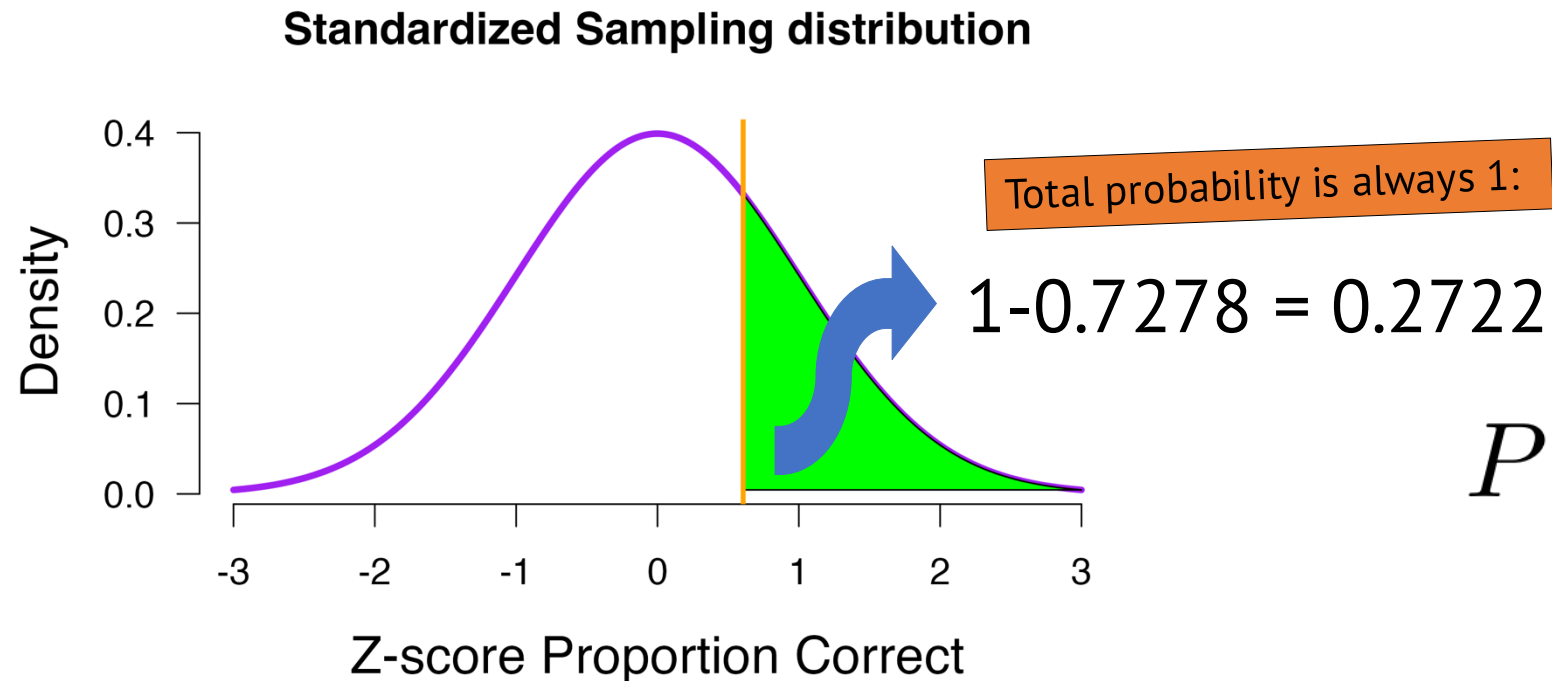
- Or use the mean and sd directly:

- = NORM.DIST(0,7368; 0,7; 0,0607; TRUE) = 0.7278



$$z = \frac{0.7368 - 0.7}{0.0607} = 0.6063$$

- = NORM.DIST(0,6063; 0; 1; TRUE) = 0.7278

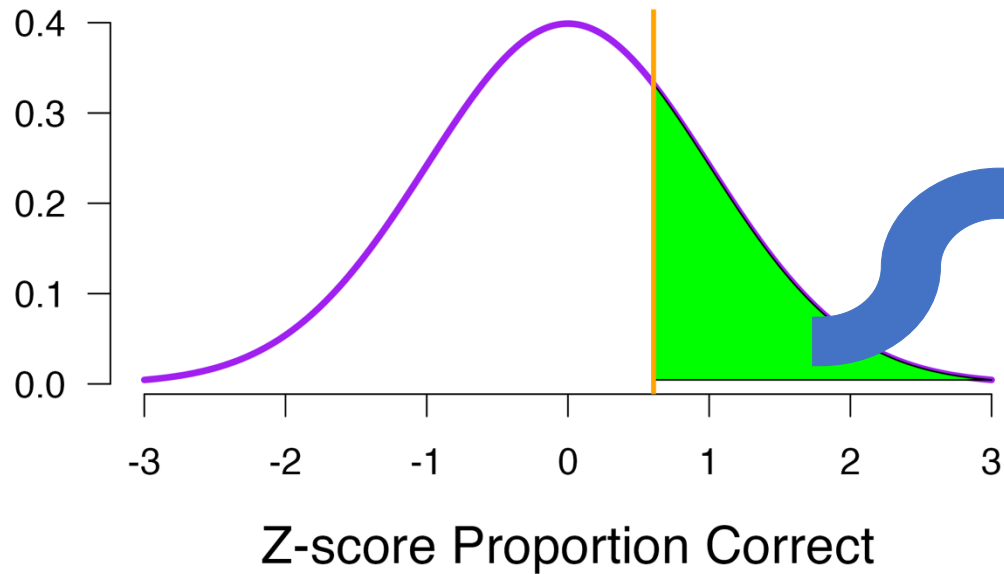


$$P(z \geq 0.6063)$$

Symmetry

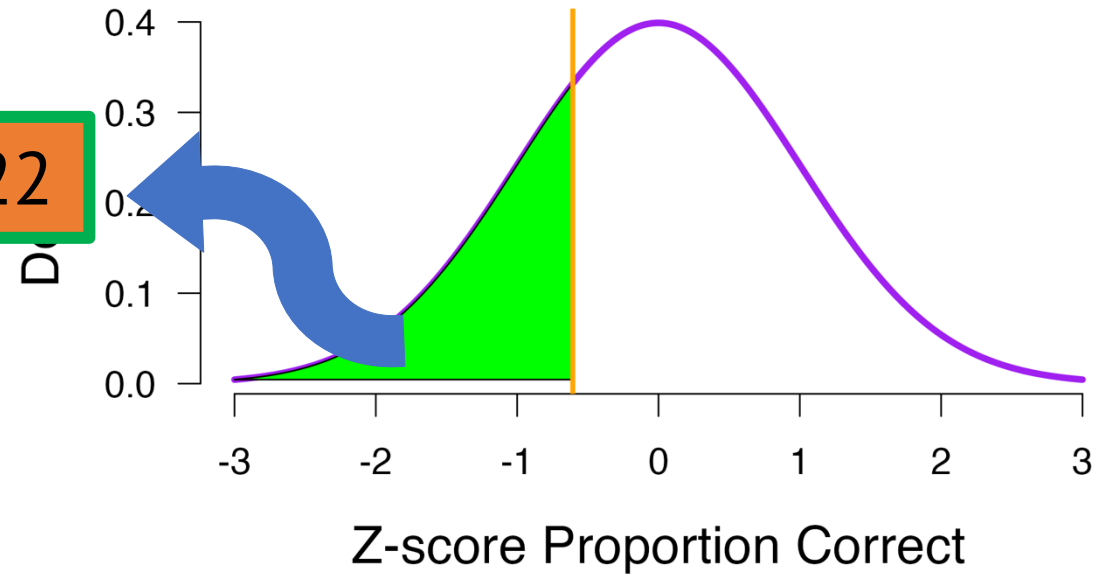
$$P(z \leq -0.6063) = P(z \geq 0.6063)$$

Standardized Sampling distribution

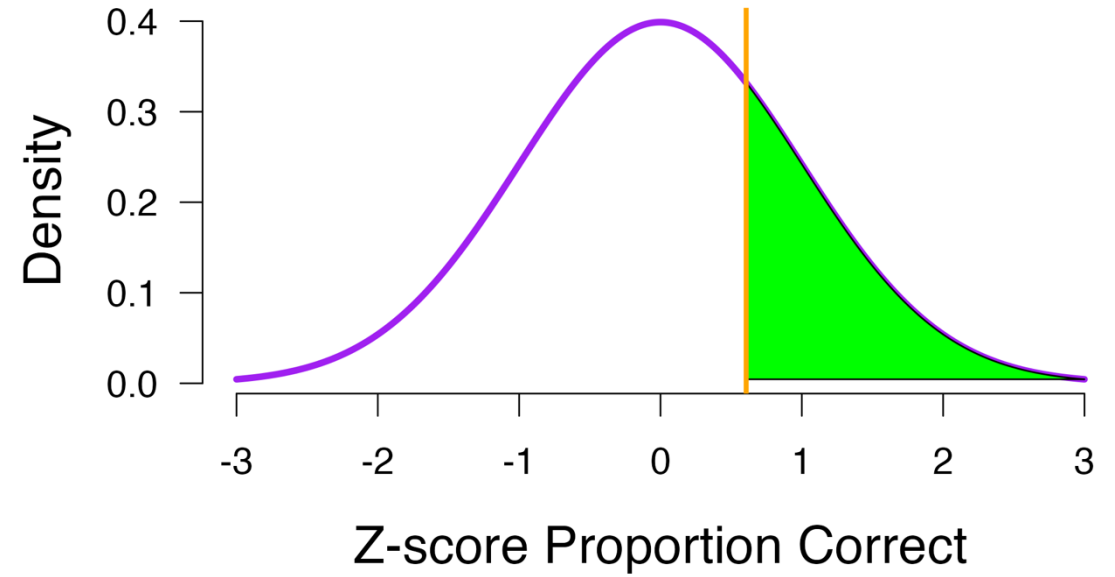


0.2722

Standardized Sampling distribution



Standardized Sampling distribution



$$P(\hat{p} \geq 0.7368 | p = 0.7) = 0.2722 \rightarrow \pm 27\%$$

So, can people taste the difference?

- If 70% of people can, 42/57 or more is not surprising (+- 27%)
- But what if everybody is just guessing ($p = 0.5$)?
- → *In hypothesis testing, we look at the probability of the observed data/statistic, given a hypothetical value of the parameter (block 3)*

Overview of Today

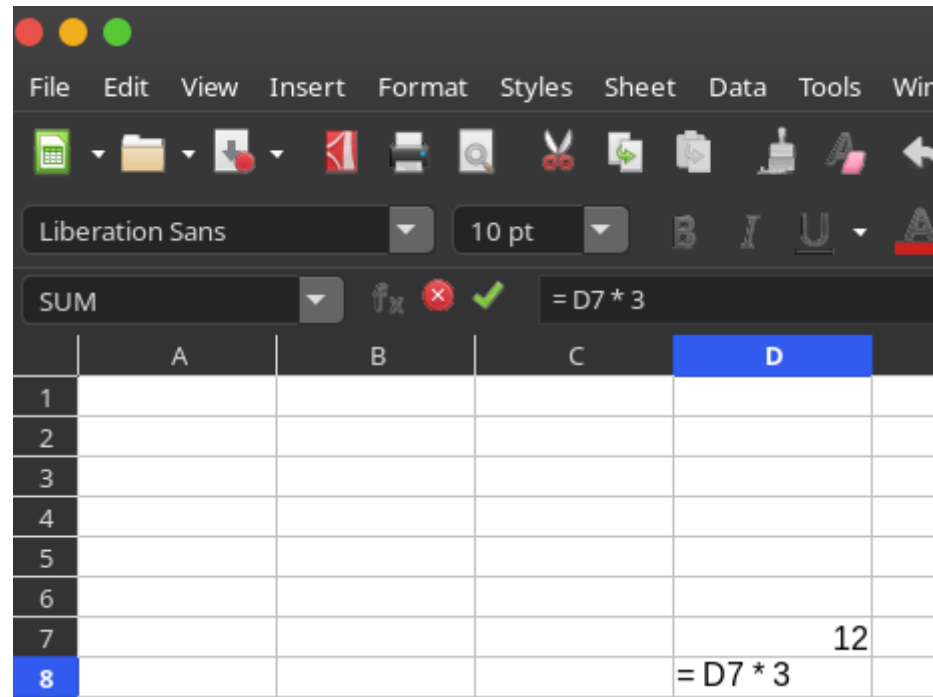
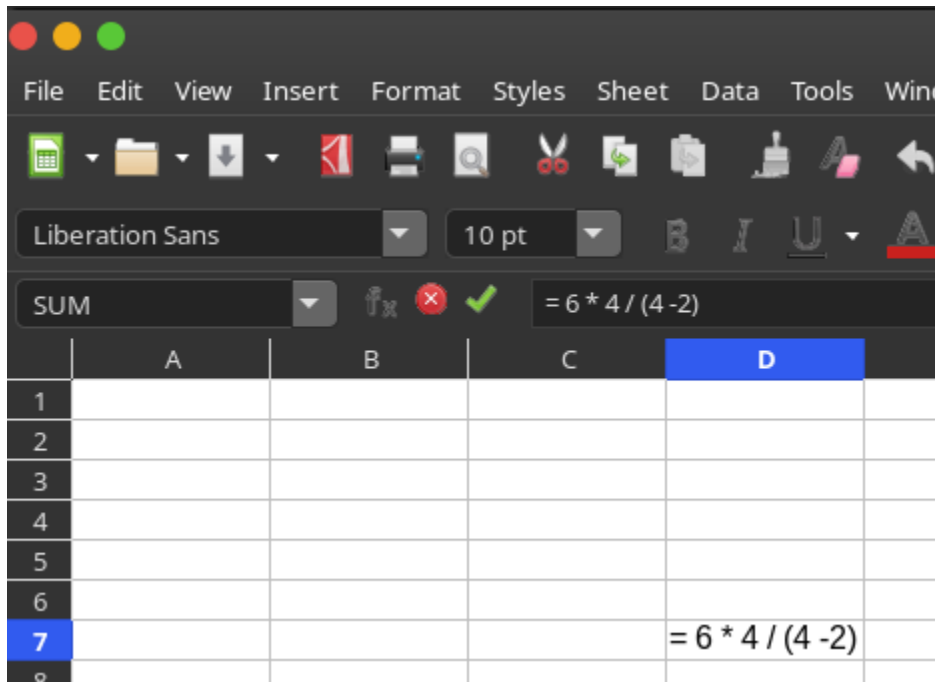
1. Population distribution vs data distribution
2. Sampling distribution
 1. Distribution of a quantitative variable
 2. Distribution of a binary categorical variable
3. Central Limit Theorem
4. **Excel & Demo**
5. Recap
 - Next time
 - Example exam questions

Excel commands for calculations:



Regular numeric operations:
 $= 2,3 + 3 - 7 * \text{SQRT}(10) / 4^2$

→ Paste these commands into any cell and press enter!



→ 36

Excel commands for calculations:

Regular numeric operations:
= 2,3 + 3 - 7 * SQRT(10) / 4^2

Microsoft office is available for free [here](#), but you can also take a look at the open-source (/free) LibreOffice!

<https://www.libreoffice.org/>



The Dress...

https://en.wikipedia.org/wiki/The_dress

Special issue of Journal of Vision:

<https://jov.arvojournals.org/SS/thedress.aspx>



The Dress...



<https://maglit.me/rms-sampling-q>

The Dress...



<https://researchmethodsuva.shinyapps.io/samplingDist/>

Overview of Today

1. Population distribution vs data distribution
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 - Example exam questions

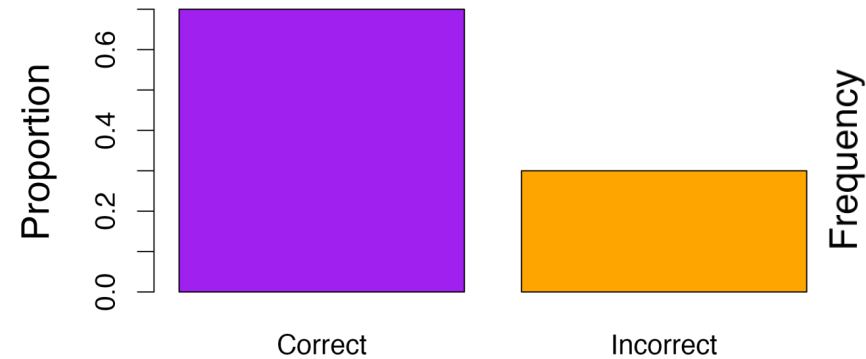
Recap

Sample (n=100)

Sampling distribution

Binomial distribution, divided by n
 $n=100, p=0.7$

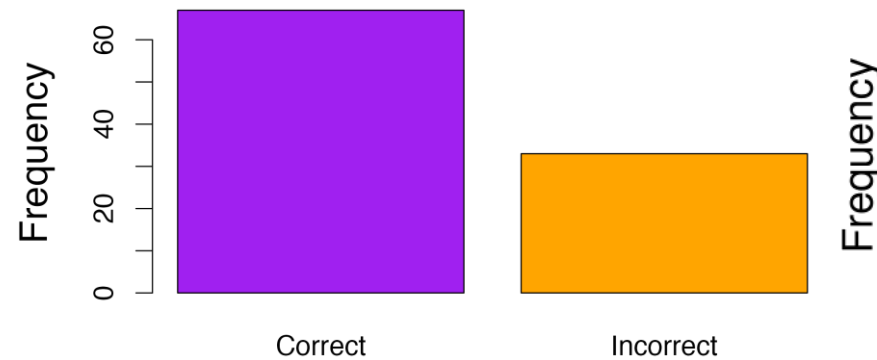
Population Distribution



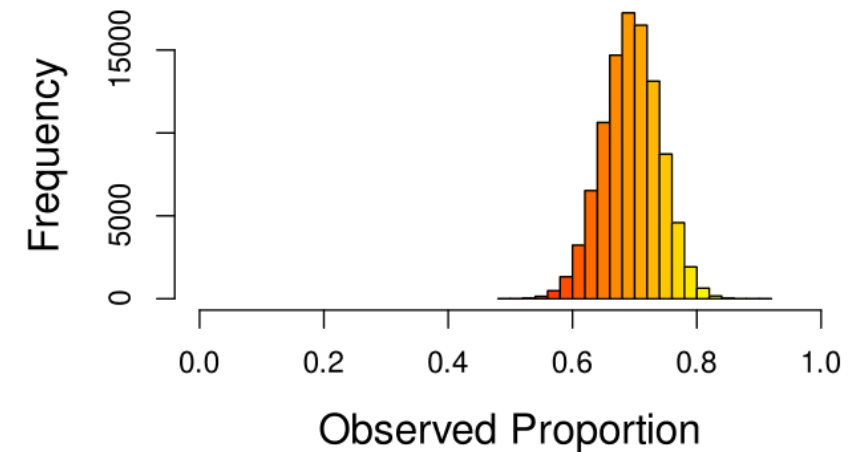
$p = 0.7$

Data Distribution

$\hat{p} = 0.67$



$\hat{p} = 0.67$

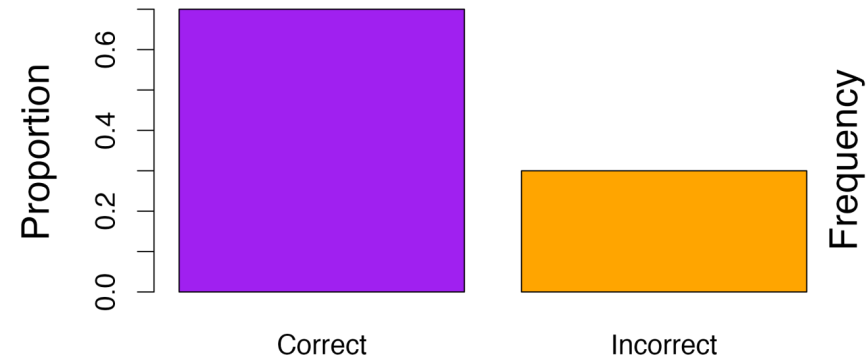


Sampling distribution: What values can you expect of your statistic, if you would repeat an experiment?

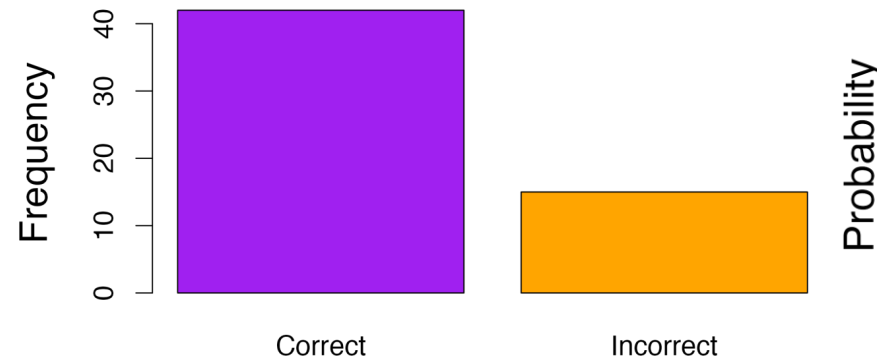
Recap

Sample (n=57)

Population Distribution



Data Distribution



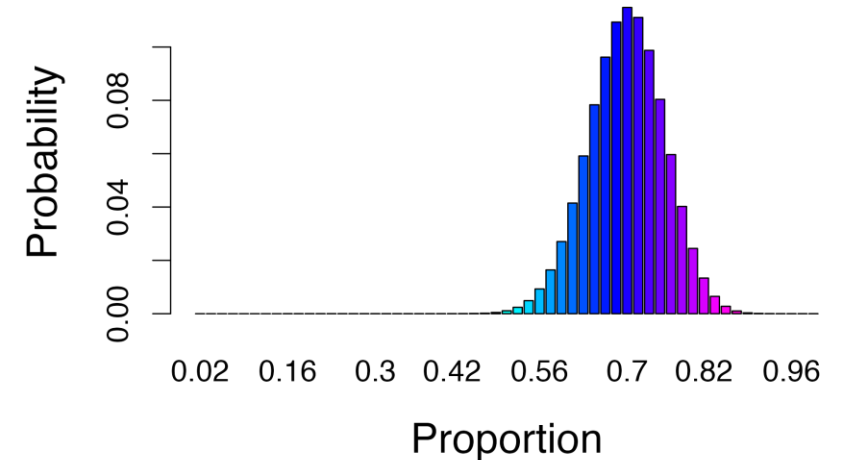
$p = 0.7$

$\hat{p} = 0.74$

Behavior of the
variable

Sampling distribution

Binomial distribution, divided by n
 $n=57, p=0.7$



Behavior of the
statistic
(on x-axis)

Recap

- Each *statistic* has a sampling distribution
- Based the sampling distribution, we can say something about the population
 - For the mean, the sampling distribution is the normal distribution
 - For the proportion, the sampling distribution is the binomial distribution divided by n
- OR the other way around: we assume/know/hypothesize the sampling distribution, so we can say something about the sample
 - Later in the course we use the sampling distribution to test hypotheses

Next Time

- This lecture: Confidence in parameter *estimates* (i.e., statistics)
- Next week: Using this confidence to draw inferences
- Problem is that we have *not observed a sampling distribution*, but *only one sample*
- → *Confidence interval*

Overview of Today

1. Population distribution vs data distribution
2. Sampling distribution
 1. Distribution of a quantitative variable
 2. Distribution of a binary categorical variable
3. Central Limit Theorem
4. Excel & Demo
5. Recap
 - Next time
 - **Example exam questions**

Example Exam Question 1

- IQ is normally distributed with a mean (μ) of 100 and a standard deviation (σ) of 15.
 - If you would measure IQ in 100 random people, what is the probability of a sample mean of 103 or more?
- a) 0.02
- b) 0.12
- c) 0.42

Solution

- Given the sample size and the Central Limit Theorem, you can assume that the sampling distribution is normally distributed with $\mu=100$ and $SE = \frac{15}{\sqrt{100}} = 1.5$

- From this, you can compute the z-score, which indicates the probability:

$$P(\bar{x} > 103) = P\left(\frac{\bar{x} - \mu}{\sigma} > \frac{103 - 100}{1.5}\right)$$

(Because total probability is always 1)

= NORM.DIST(2; 0; 1; TRUE) = 0.9772, therefore $P(z > 2) = 1 - 0.9772 = 0.0228$

- Alternative: $P(z > 2) = P(z < -2)$ ← Because of symmetry

= NORM.DIST(103; 100; 1.5; TRUE) also possible -> without conversion to z

Example Exam Question 2

- An old study showed that 4% of Dutch children under the age of 16 have ADHD. A researcher wants to understand whether that percentage is still accurate. She takes a random sample of 500 children from the appropriate population. 5.5% of the children in the sample have ADHD.
 - What is the probability of finding a percentage of 5.5% or more in a sample, if the population percentage is 4%?
- a) 0.04
 - b) 0.14
 - c) 0.47

Solution

We know that the sampling distribution is binomially distributed with $p = 0.04$ and

$$SE = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.04(1-0.04)}{500}} = 0.0088$$

- Because of the sample size and the Central Limit Theorem, we can assume that the sampling distribution of the proportion is approximately normal

- $np \geq 15$

$$500 * 0.04 = 20$$

- $n(1-p) \geq 15$

$$500 * (1 - 0.04) = 480$$

Solution

- We can calculate the z-score and look it up in Excel:
- $Z\text{-score} = (0.055 - 0.04) / 0.0088 = 1.7045$

Solution

- From this, you can compute the z-score, which indicates the probability:

$$P(\hat{p} > 0.055) = P\left(\frac{\hat{p} - \mu}{SE} > \frac{0.055 - 0.04}{0.0088}\right) = P(z > 1.7045) = 1 - P(z < 1.7045)$$

(Because total probability is always 1)

- $P(z < 1.7045) = \text{NORM.DIST}(1.7045; 0; 1; \text{TRUE}) = 0.9559$
 - therefore $P(z > 1.7045) = 1 - 0.9559 = 0.0441$
- Alternative: $P(z > 1.7045) = P(z < -1.7045) \rightarrow$ Because of symmetry
 - $= \text{NORM.DIST}(-1.7045; 0; 1; \text{TRUE}) = 0.0441$

Solution

- You can also skip the converting to z-scores and plug in mu and se directly into your Excel command:
 - = NORM.DIST(0,055; 0,04; 0,0088; TRUE) = 0.9559
 - = 1 - NORM.DIST(0,055; 0,04; 0,0088; TRUE) = 0.0441

Questions?

Thank you for your attention



THE PROBLEM WITH
AVERAGING STAR RATINGS

Highlighted exercises from the book

- 7.40
- 7.46
- 7.50

→ try yourself first, then check
next slides for answers

7.40

a) $n = 50$, $p = 0.7$; mean = p , sd = $\sqrt{(0.7 * 0.3) / 50} = 0.0648$

b) It will approximate the normal distribution, since

- $np = 50 * 0.7 = 35$
- $n(1-p) = 50 * 0.3 = 15$
- Both are greater than or equal to 15

c) See next slide

$$\sqrt{\frac{p(1-p)}{n}}$$

7.40

a)

b)

c) Convert the observation (0.6) to z-score, using the sampling distribution mean and standard deviation:

- $(0.6 - 0.7) / 0.0648 = -1.5432$
- Then use Excel to get the probability of this observation or lower (i.e., the area under the curve to the left of z):
- $= \text{NORM.DIST}(-1.5432; 0; 1; \text{TRUE}) = 0.0614$
- So if the population mean is 0.7, observing 0.6 or less (with 50 questions) has a probability of +/- 6%, which is pretty low/surprising (meaning something might have gone wrong in your exam preparations!)

7.46

- a) The mean is 900, and standard deviation 300, for the population (see the symbols – these indicate the parameters)
- b) The data statistics are also reported in the story: mean is 980 and standard deviation is 276 – again, these are about the **variable**, so this is about the spread of the daily sales.
- c) See next slide

7.46

a)

b)

c) Mean and standard deviation of sampling distribution are about the **statistic**. The mean is equal to the population mean, and the standard deviation (i.e., standard error) is equal to the population sd divided by $\sqrt{n}$. The average is computed over 7 days, so the SE is equal to $300 / \sqrt{7} = 113.39$. The SE describes how much the **mean** varies if we were to keep taking 7-day samples and compute the mean for those 7 days.

7.50

a) Since this is about the z-score for **a single value** of PDI (and not a statistic), we use the data standard deviation instead of the SE:

$$\text{Z-score: } (90 - 100) / 15 = -0.6667$$

b) Now we are talking about the **mean** PDI, so we use the SE for the z-score calculation:

$$\text{Z-score: } (90 - 100) / (15 / \sqrt{225}) = -10$$

c) See next slide

7.50

This wall of text is just meant for intuition – your answer does not need to completely overlap

a)

b)

c) The z-score for the mean of 225 observations is -10, whereas the z-score for a single observation is -0.6667. In the latter case, you can see the score of 90 as the mean PDI for a sample of 1 (so the standard error is equal to sigma). For a sample size of 1, the sampling distribution is much more wide than for a sample size of 225, since these values will fluctuate a lot more. It is therefore much more unlikely to observe a mean of 90 based on 225 participants. (extra intuition: to observe a mean of 90, a lot of those 225 need to score even lower than that 90, which is more unlikely than scoring 90)



Source: Pixabay.org